Mikroelektronische Schaltungen und Systeme

Lect.8
Stability

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Mikroelektronische Schaltungen und Systeme

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Introduction

- We have seen that amplifiers with second-order response can result in an oscillatory behavior,
- We would like to take a deeper look into this phenomena,
- Understand unstable behavior, how to assess it and how to eliminate it.

2nd Order Amplifiers

- The second-order system will be described with the characteristic equation in the denominator

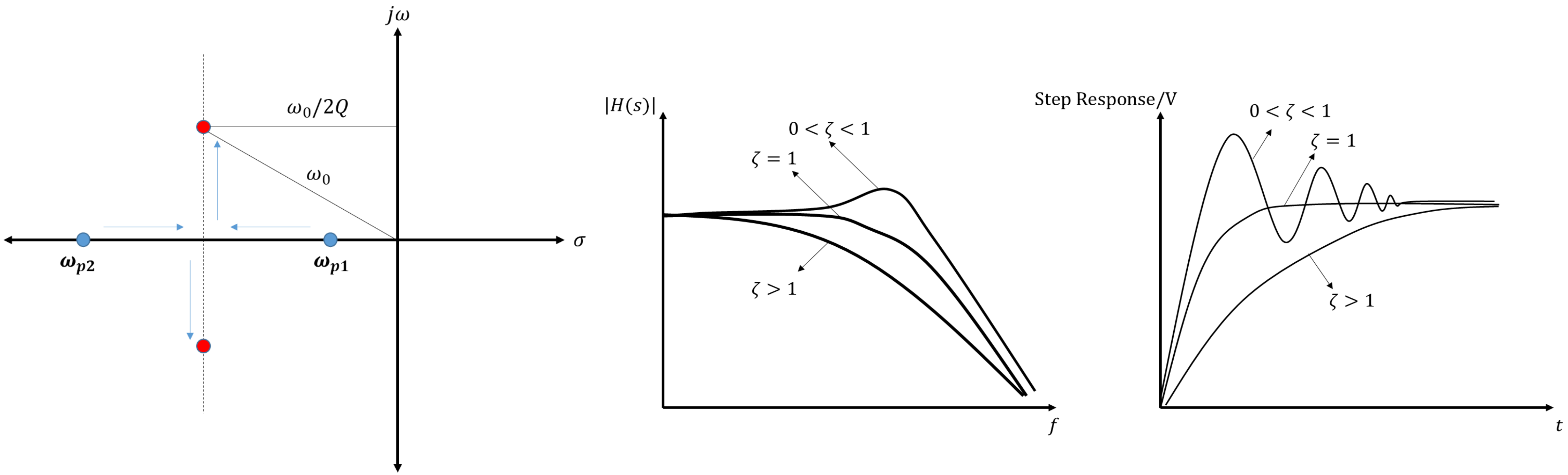
$$\boldsymbol{s^2 + s\frac{\omega_0}{Q} + \omega_0^2 = 0}$$

- In terms of the time constants

$$1 + s(\tau_1 + \tau_2) + s^2\tau_1\tau_2$$

2nd Order Amplifiers

- The behavior we have previously seen



- is observed, when the roots of the characteristic function are complex conjugates

2nd Order Amplifiers

- Quadratic equation:

$$1 + s(\tau_1 + \tau_2) + s^2\tau_1\tau_2$$

$$\mathrm{c} + sb + s^2 a$$

- The solution of the quadratic equation:

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- s here of course being the Laplace parameter with $s = \sigma + j\omega$ ($\sigma = 0$ is steady state)
- The roots will be complex conjugate if $b^2 < 4ac$

2nd Order Amplifiers

- In terms of circuit parameters:

$$1 + s(\tau_1 + \tau_2) + s^2\tau_1\tau_2$$

$$a = \tau_1\tau_2$$
$$b = \tau_1 + \tau_2$$
$$c = 1$$

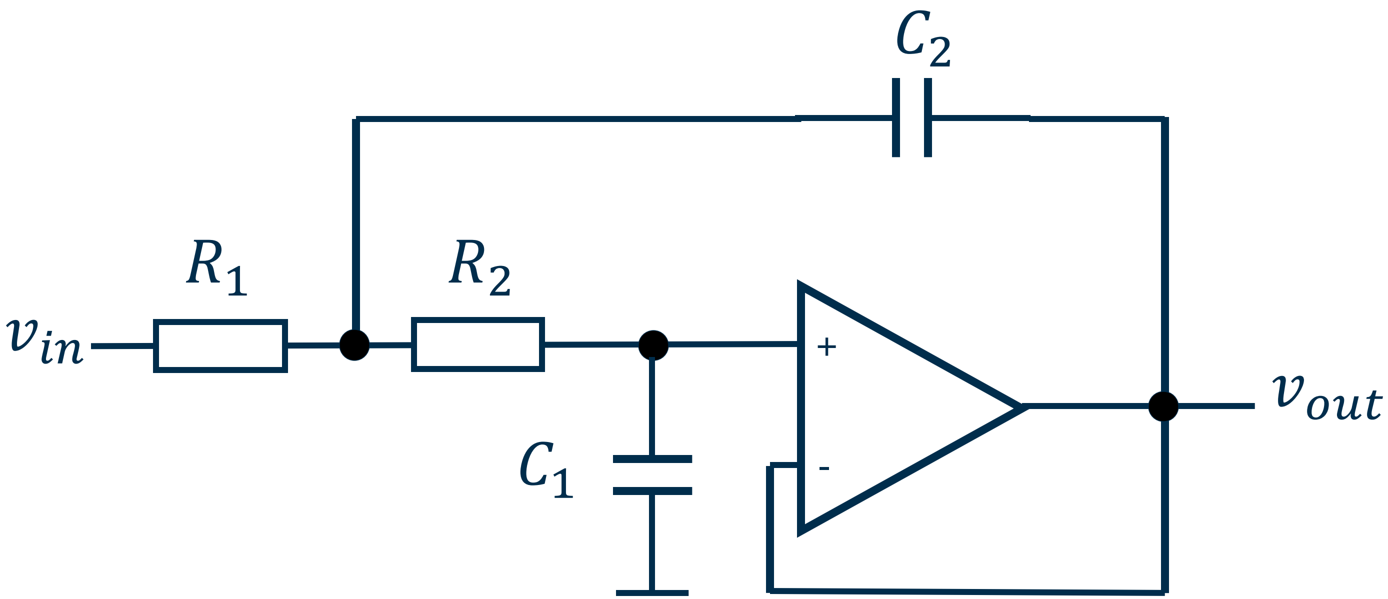
- For complex roots:

$$(\tau_1 + \tau_2)^2 < 4\tau_1\tau_2$$

- This typically requires some kind of feedback, or interaction between reactive components beyond a simple second-order response

Sallen-Key Filter

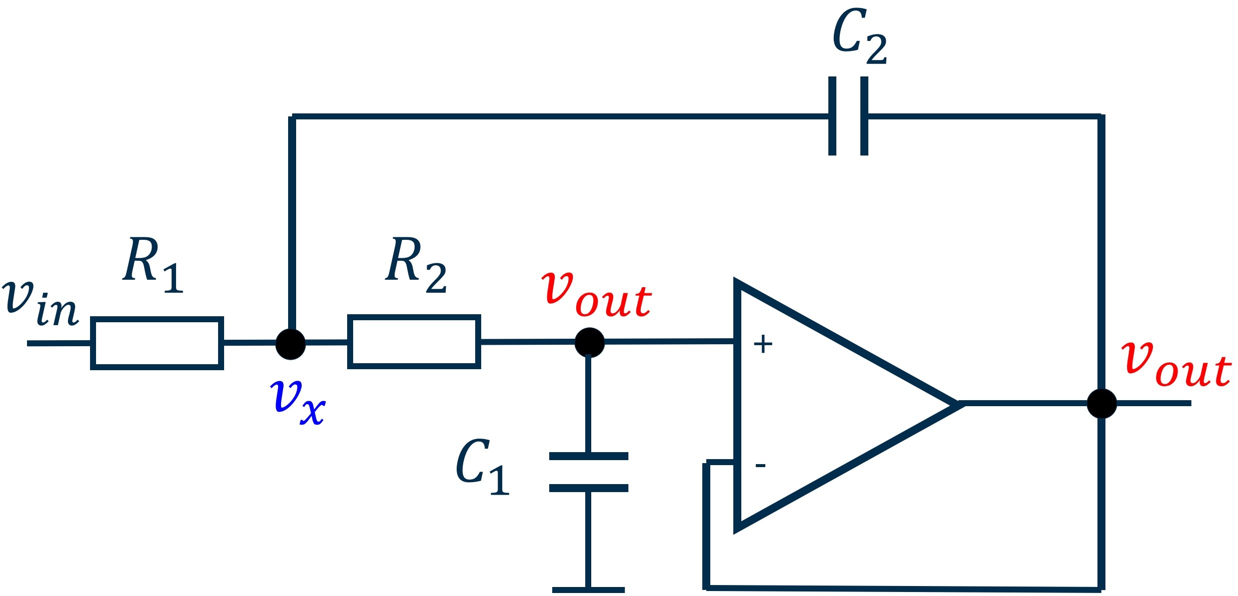
- We can observe this effect with a well-know circuit architecture: the Sallen-Key filter



- We will focus on asymptotic transfer function H_∞. In this example this is quite accurate, considering $H_0 = 0$ for output open and a high OPAMP gain

Sallen-Key Filter

- In the asymptotic case (ideal OPAMP):



Sallen-Key Filter

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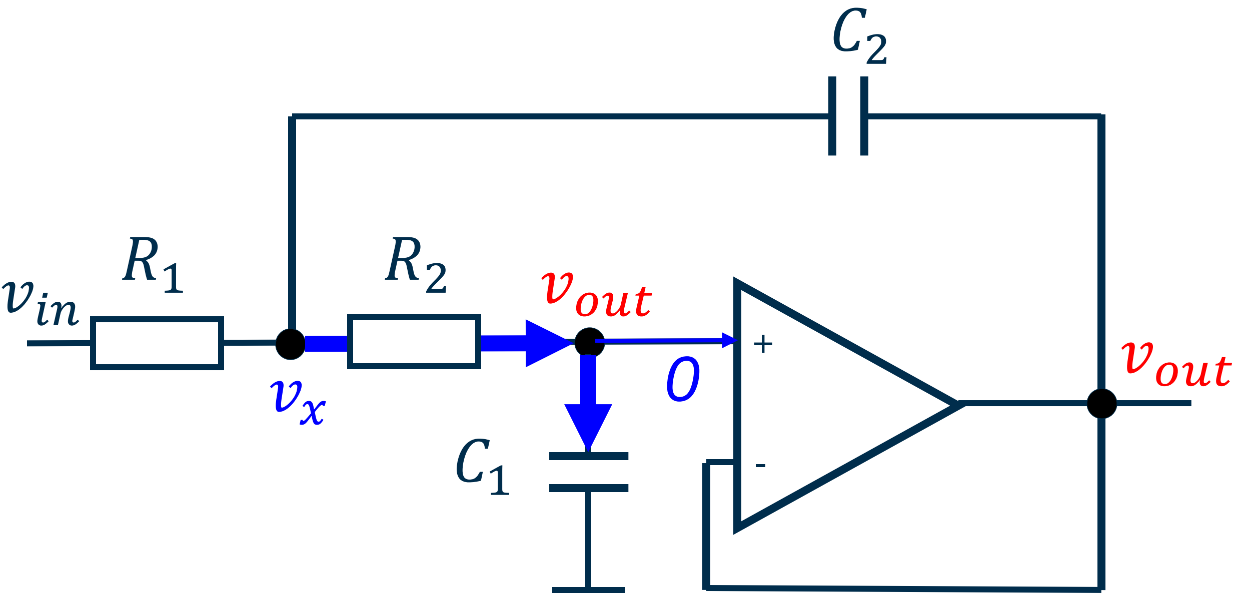
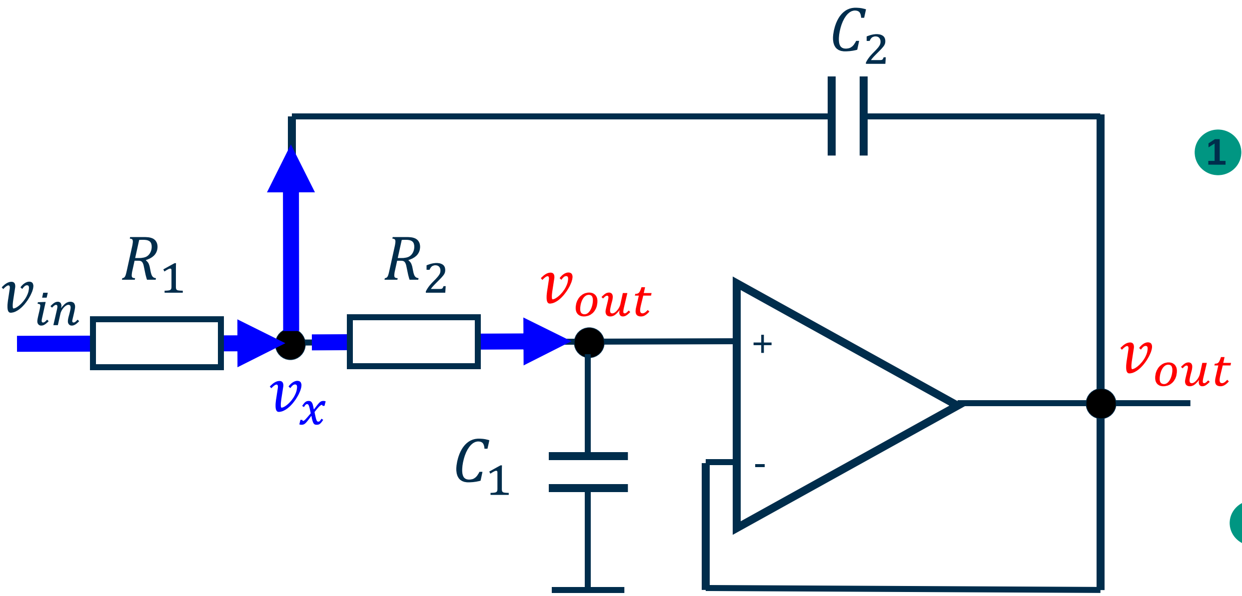


$$\frac{v_x - v_{out}}{R_2} = sC_1 v_{out} \rightarrow v_x = v_{out}\,(1 + sR_2C_1)$$

Sallen-Key Filter

- In the asymptotic case (ideal OPAMP):

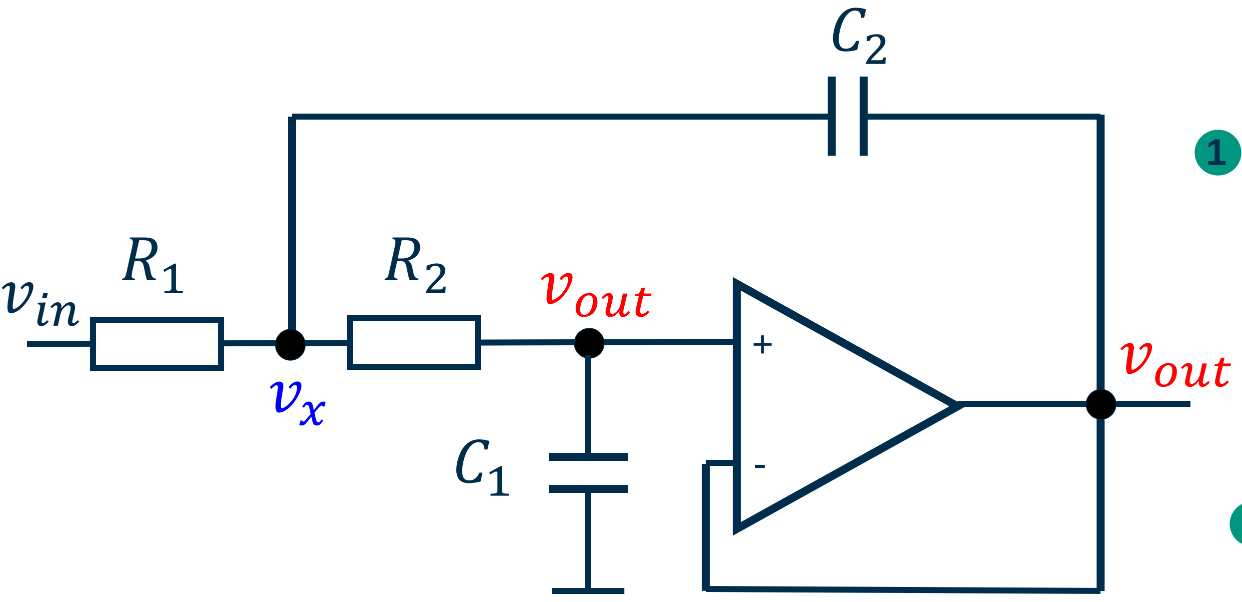


1. $$\frac{v_x - v_{out}}{R_2} = sC_1 v_{out} \rightarrow v_x = v_{out}\,(1 + sR_2C_1)$$

2. $$\frac{v_{in} - v_x}{R_1} = (v_x - v_{out})sC_2 + \frac{v_x - v_{out}}{R_2}$$

Sallen-Key Filter

- In the asymptotic case (ideal OPAMP):



1 $$\frac{v_x - v_{out}}{R_2} = sC_1 v_{out} \rightarrow v_x = v_{out}\,(1 + sR_2C_1)$$

2 $$\frac{v_{in} - v_x}{R_1} = (v_x - v_{out})sC_2 + \frac{v_x - v_{out}}{R_2}$$

$$v_{in} - v_{out}\,(1 + sR_2C_1) = v_{out}s^2R_2R_1C_1C_2 + v_{out}sR_1C_1$$

$$\boxed{\frac{v_{out}}{v_{in}} = \frac{1}{s^2R_2R_1C_2C_1 + s(R_1C_1 + R_2C_1) + 1}}$$

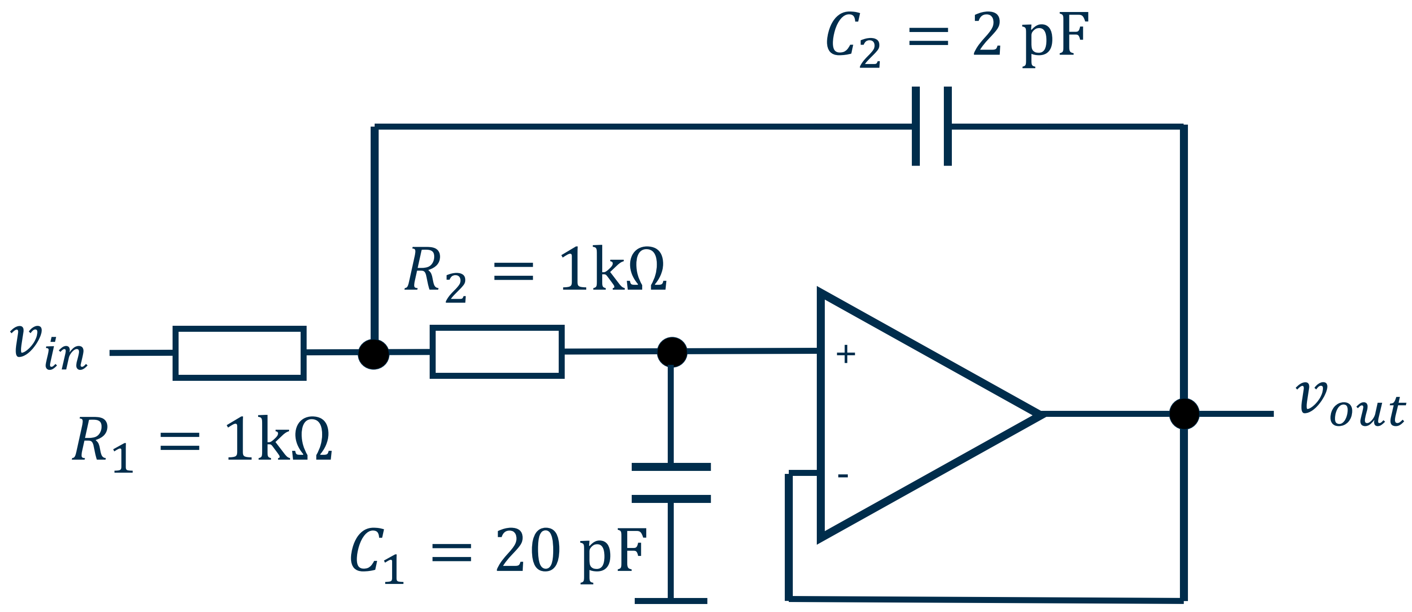
Sallen-Key Filter

- Here, we see that C_2 allows us to adjust *a* and *b* coefficients independently

$$\frac{v_{out}}{v_{in}} = \frac{1}{s^2 R_2 R_1 {\color{red}C_2} C_1 + s(R_1 C_1 + R_2 C_1) + 1}$$

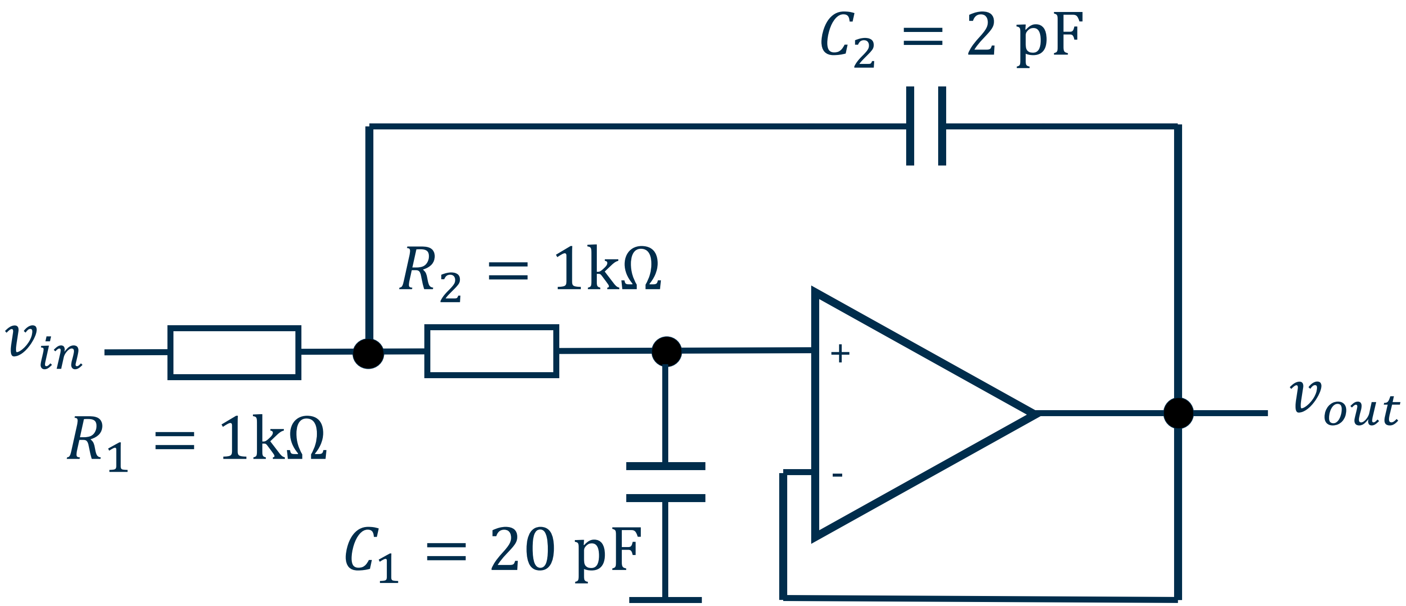
$$\frac{v_{out}}{v_{in}} = \frac{1}{s^2 + s\frac{R_1 C_1 + R_2 C_1}{R_2 R_1 C_2 C_1} + R_2 R_1 C_2 C_1}$$

Sallen-Key Filter



$$R_2 R_1 C_2 C_1 = 40\ \mathrm{ns}^2 = \frac{1}{\omega_0^2} \rightarrow f_0 = 25\ \mathrm{MHz}$$

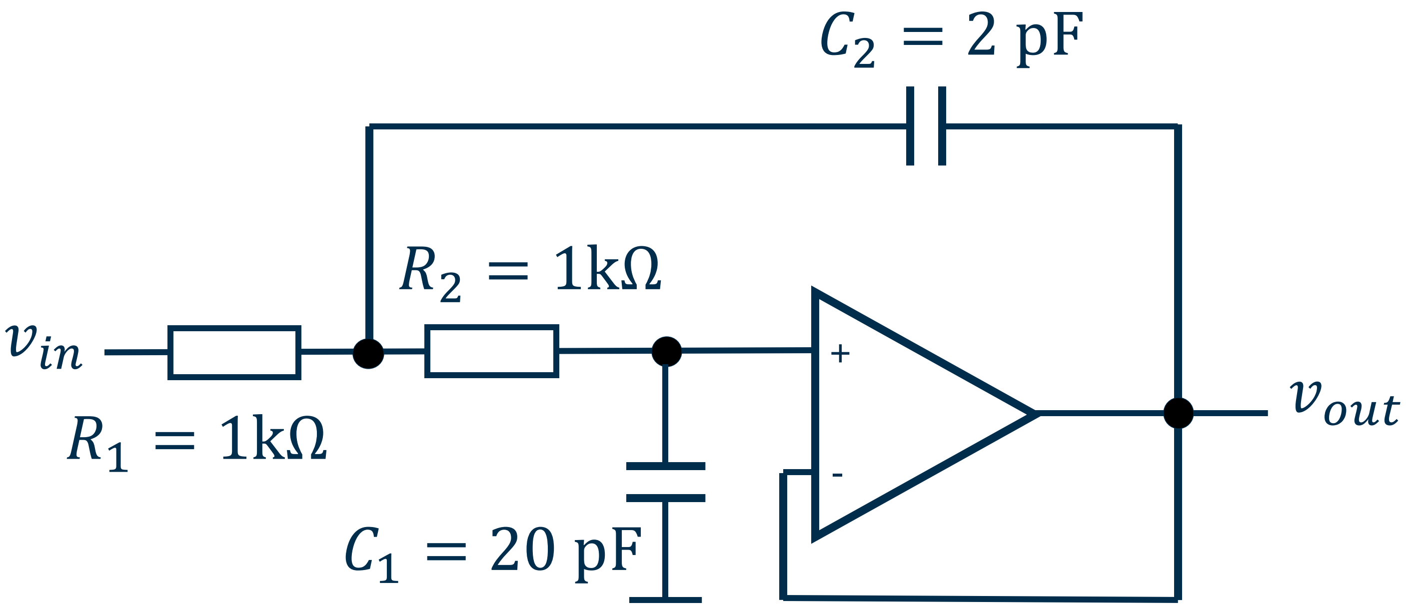
Sallen-Key Filter



$R_2 R_1 C_2 C_1 \rightarrow f_0 = 25$ MHz

$(R_1 + R_2)C_1 = b = 40$ ns $\rightarrow f_{3dB} \cong 3.98$ MHz (in case of a dominant pole)

Sallen-Key Filter

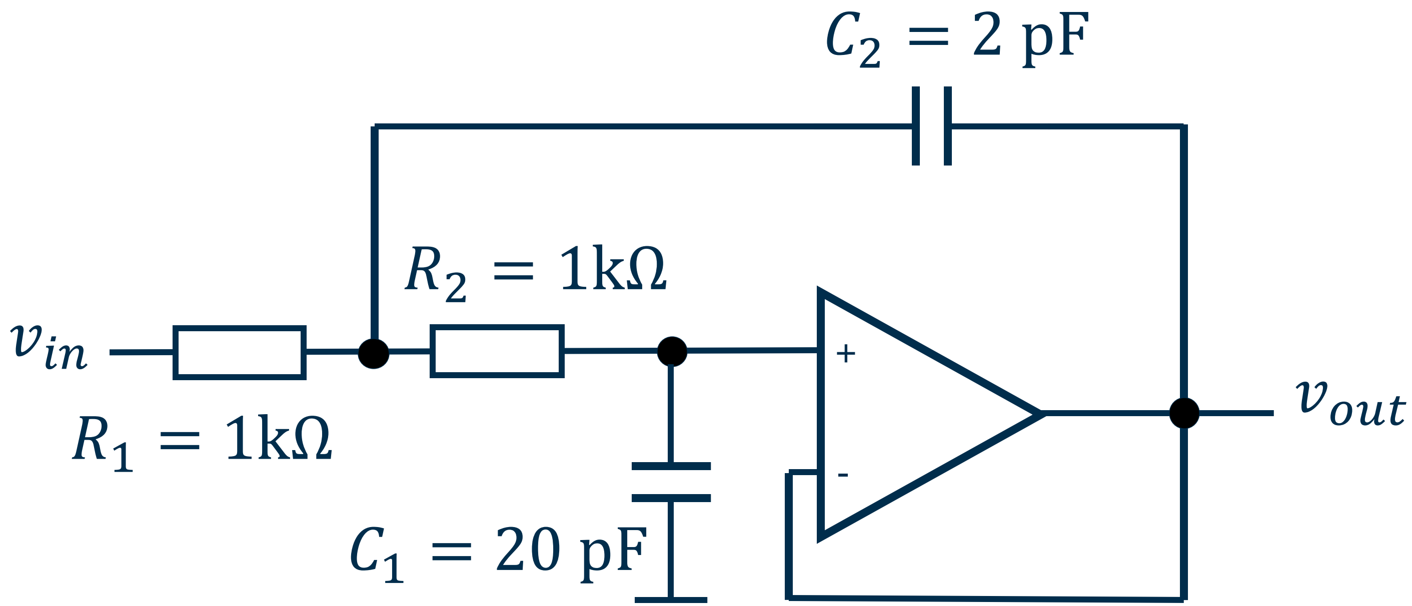


$$R_2 R_1 C_2 C_1 \rightarrow f_0 = 25\ \mathrm{MHz}$$

$$(R_1 + R_2)C_1 = b = 40\ \mathrm{ns} \rightarrow f_{3dB} \cong 3.98\ \mathrm{MHz}$$

$\tau_2 = C_2(R_1 \| R_2) = 1\ \mathrm{ns}$ -> second pole around 160 MHz

Sallen-Key Filter

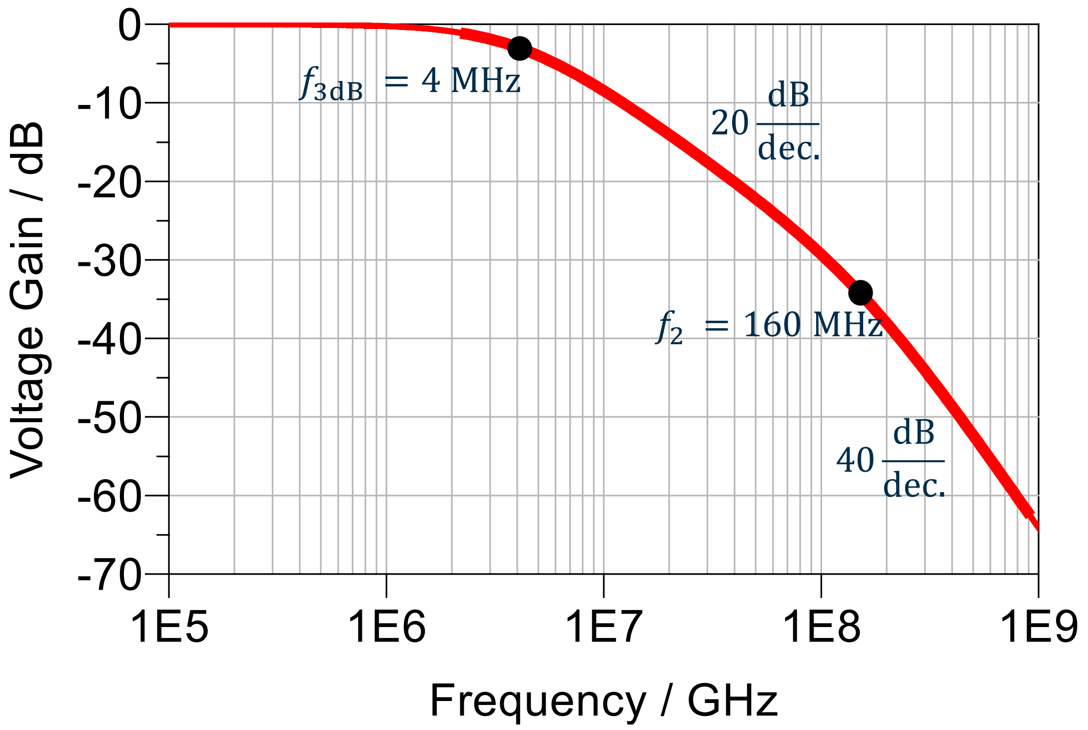


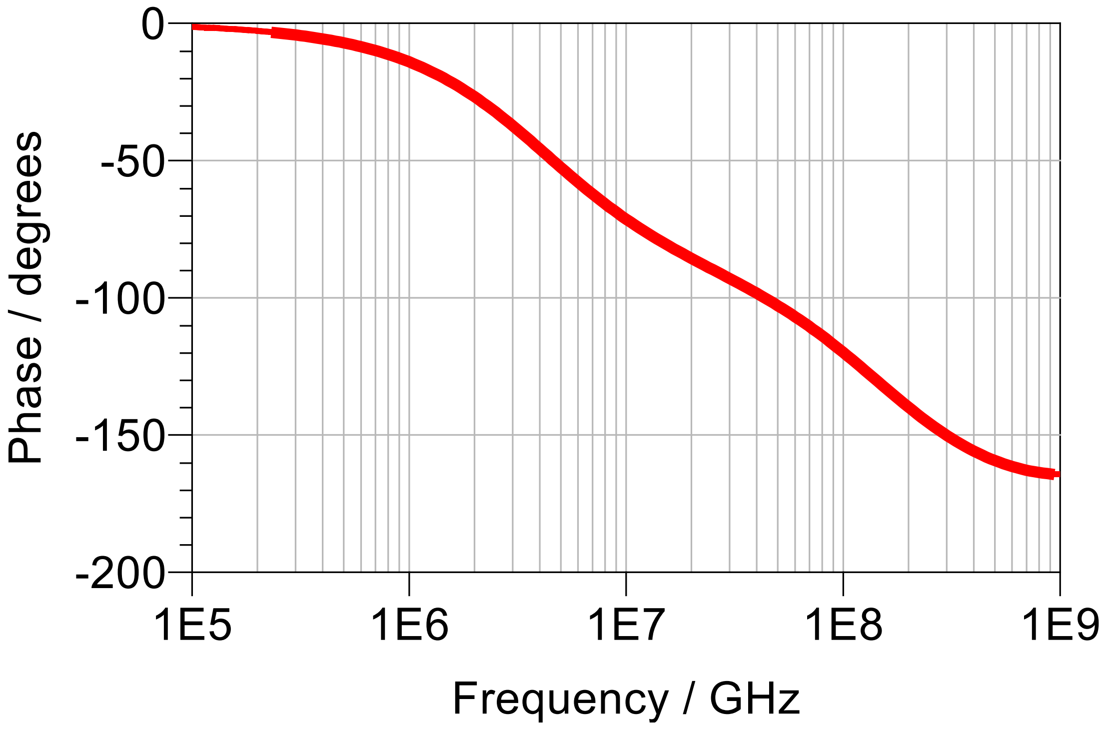
$$R_2 R_1 C_2 C_1 \rightarrow f_0 = 25\ \mathrm{MHz}$$

$$R_1 C_1 + R_2 C_1 \rightarrow f_{3dB} \cong 3.98\ \mathrm{MHz}$$

$$\frac{R_1 C_1 + R_2 C_1}{R_2 R_1 C_2 C_1} = 1\ \mathrm{Gs} = \frac{\omega_0}{Q} \rightarrow Q = 0.156 \rightarrow \zeta = 12.7$$

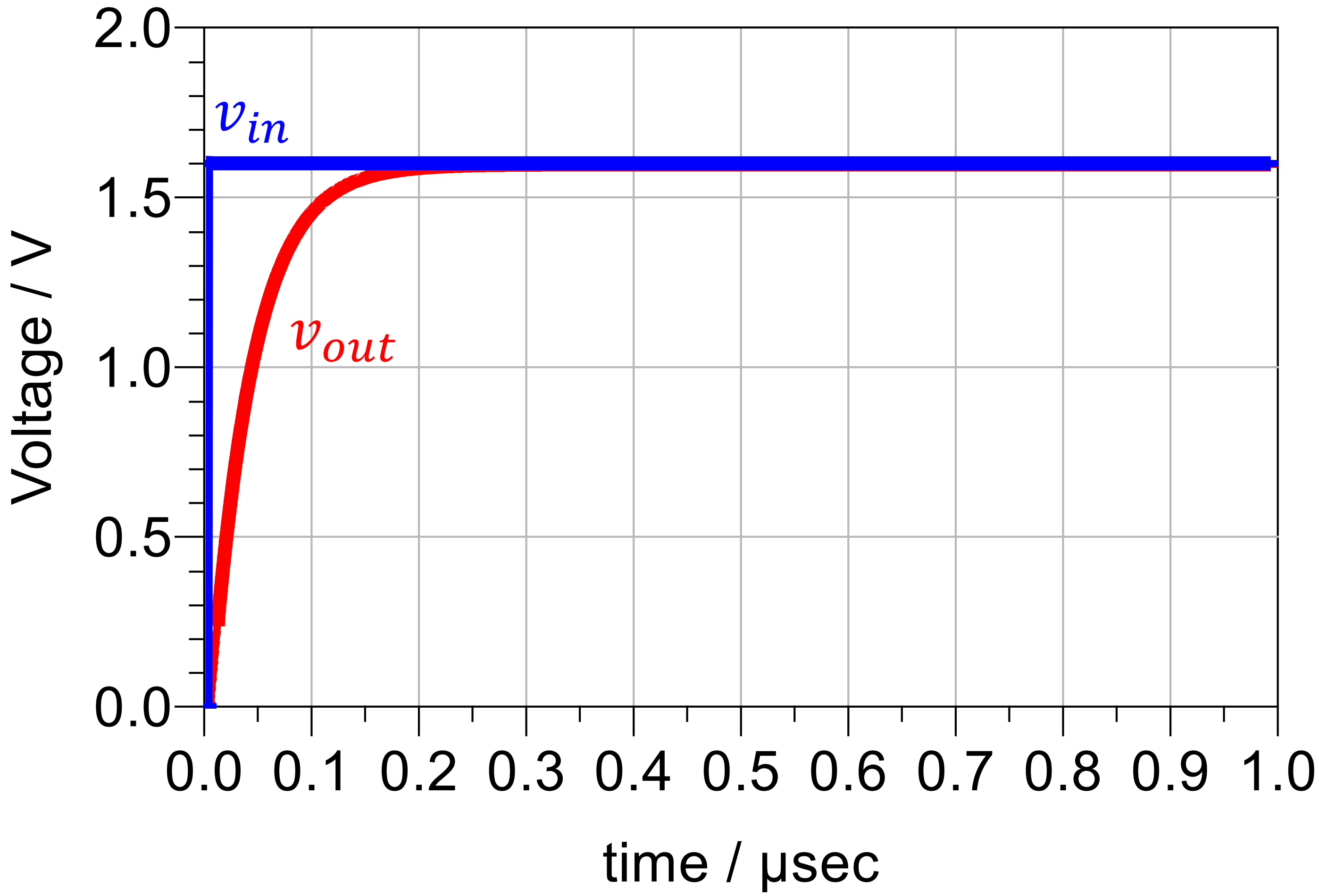
Sallen-Key Filter ($C_2 = 2\ \mathrm{pF}$)



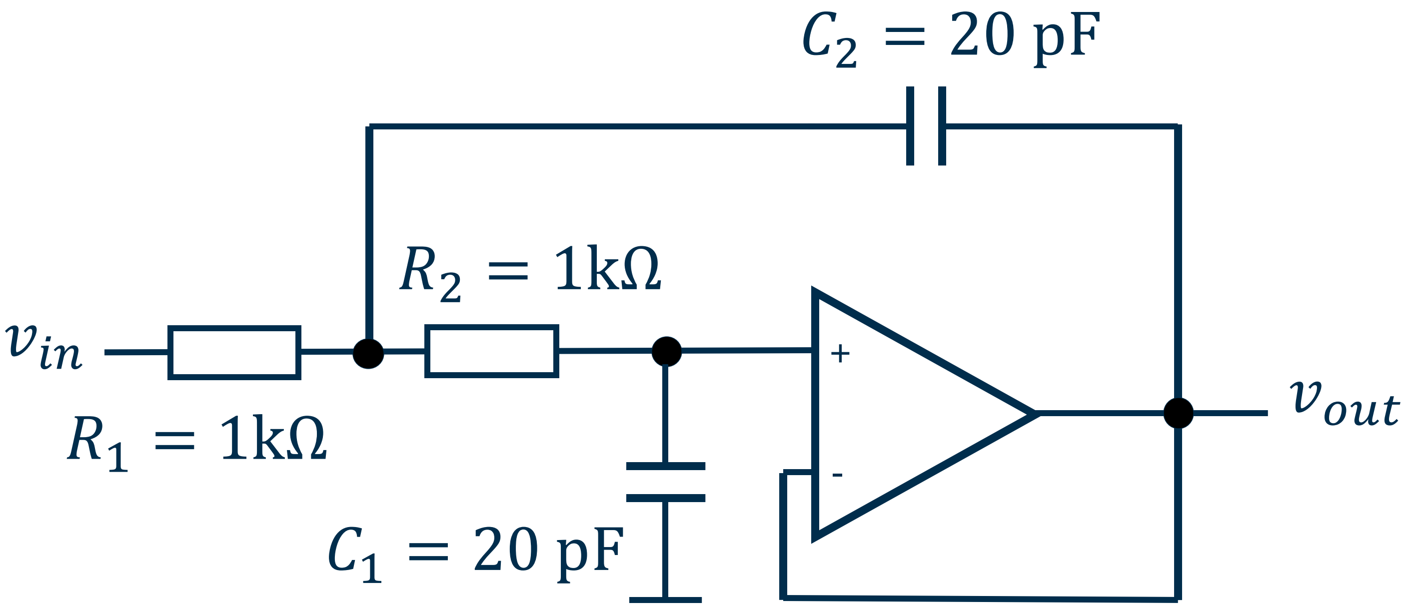


Sallen-Key Filter ($C_2 = 2$ pF)

- Step Response:



Sallen-Key Filter ($C_2 = 20\ \mathrm{pF}$)

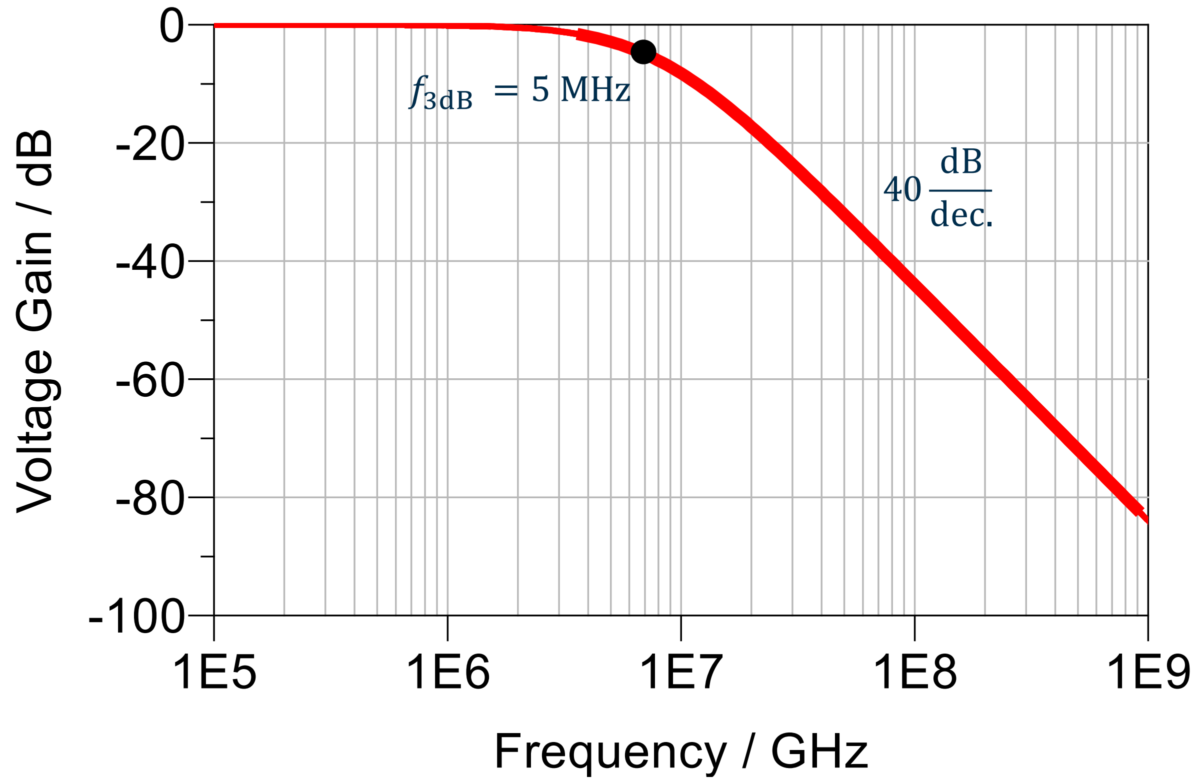


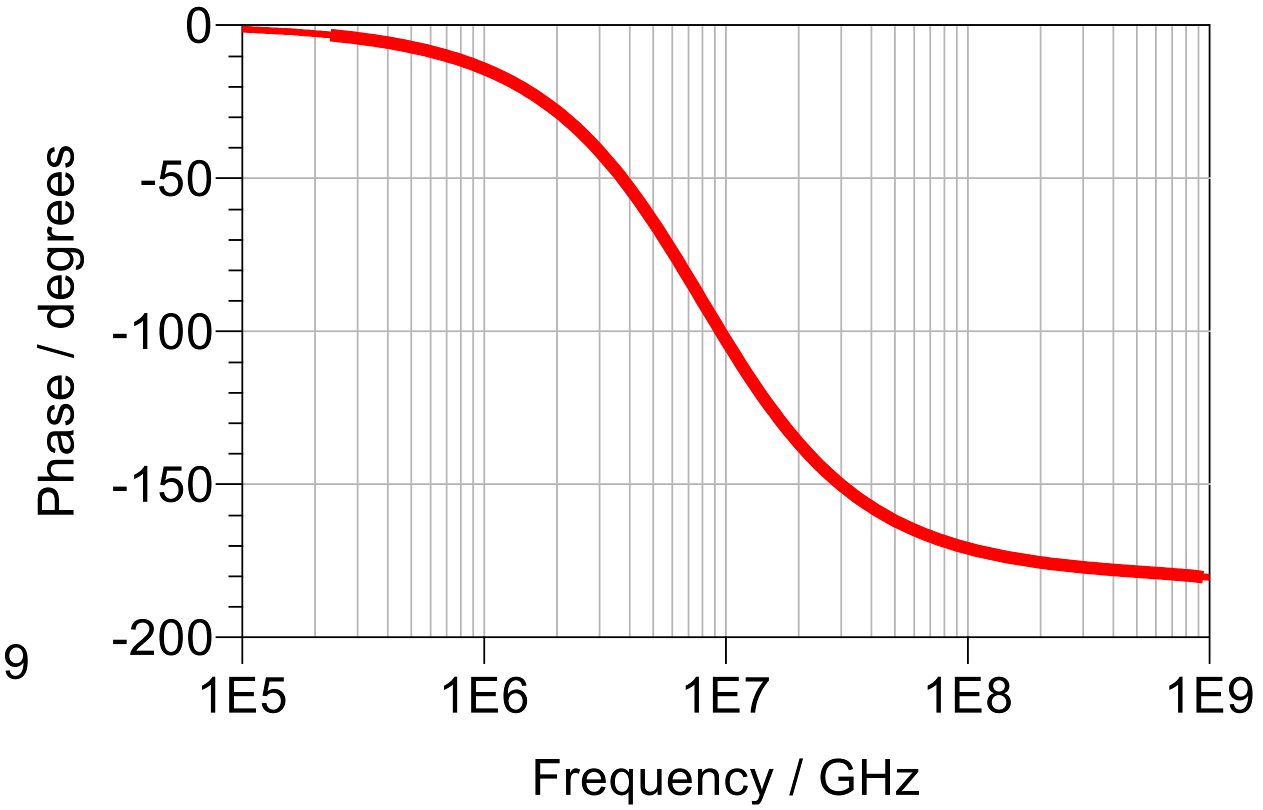
$$R_2 R_1 C_2 C_1 = 4.0 \times 10^{-16}\mathrm{s} \rightarrow \omega_0 = 50\ \mathrm{MHz}\ \rightarrow = 7.9\ \mathrm{MHz}$$

$$R_1 C_1 + R_2 C_1 = 40\ \mathrm{ns} \rightarrow f_{3dB} \neq 3.98\ \mathrm{MHz}$$

$$\frac{R_1 C_1 + R_2 C_1}{R_2 R_1 C_2 C_1} \rightarrow Q \sim 0.5\ \rightarrow \zeta \sim 1$$

Sallen-Key Filter ($C_2 = 20\ \mathrm{pF}$)





Sallen-Key Filter ($C_2 = 20\ \mathrm{pF}$)

- We rexamine the second-order transfer function

$$H(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

- If we calculate the magnitude and derive the frequency, where the magnitude is reduced by $1/\sqrt{2}$, we find:

$$\omega_{3dB} = \omega_0 \sqrt{1 - 2\zeta^2 + \sqrt{(2\zeta^2 - 1)^2 + 1}}$$

- This is only relevant for ζ approaching 1 or lower, for all cases, dominant pole approximation holds

Sallen-Key Filter ($C_2 = 20\ \mathrm{pF}$)

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$$H(s) = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

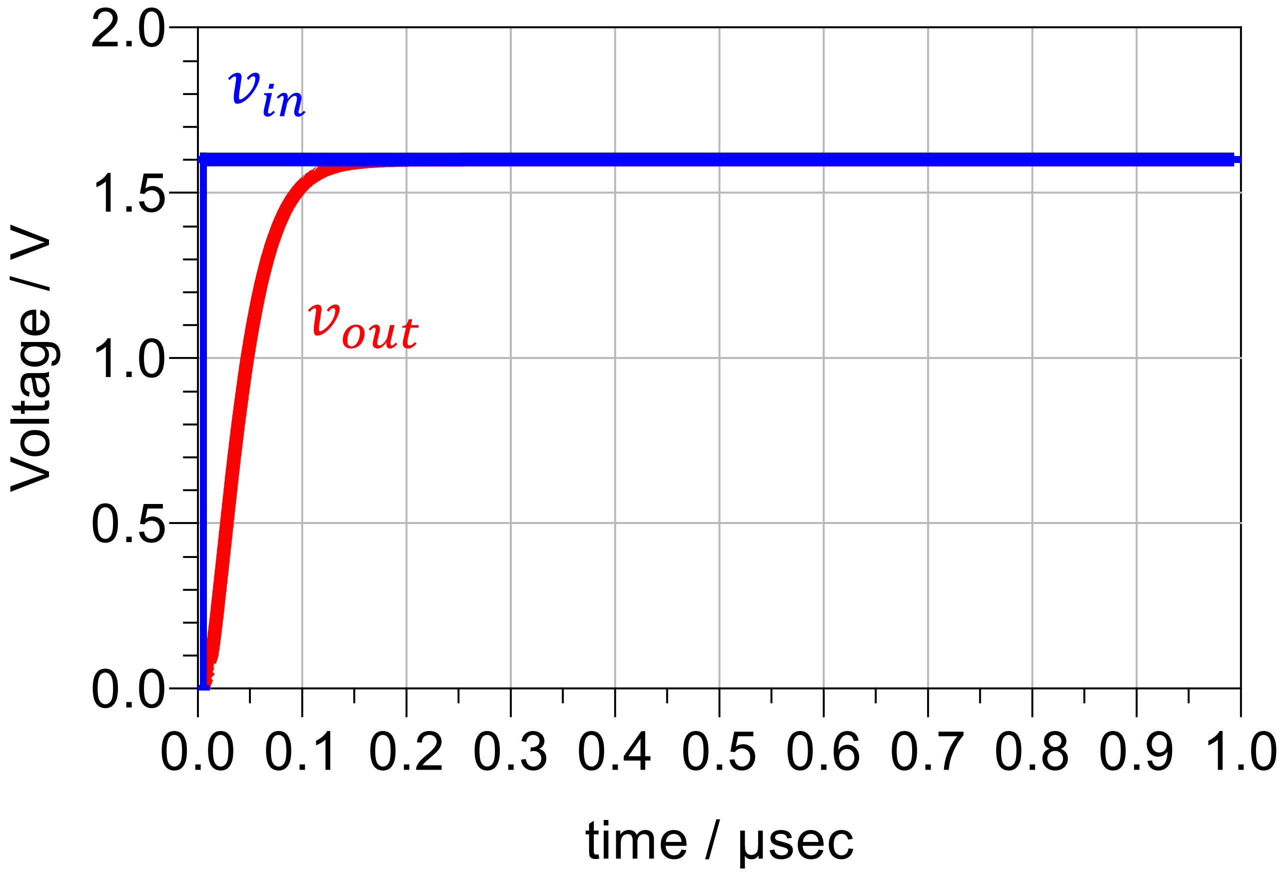
- If we calculate the magnitude and derive the frequency, where the magnitude is reduced by $1/\sqrt{2}$, we find:

$$\omega_{3dB} = \omega_0\sqrt{1 - 2\zeta^2 + \sqrt{(2\zeta^2 - 1)^2 + 1}}$$

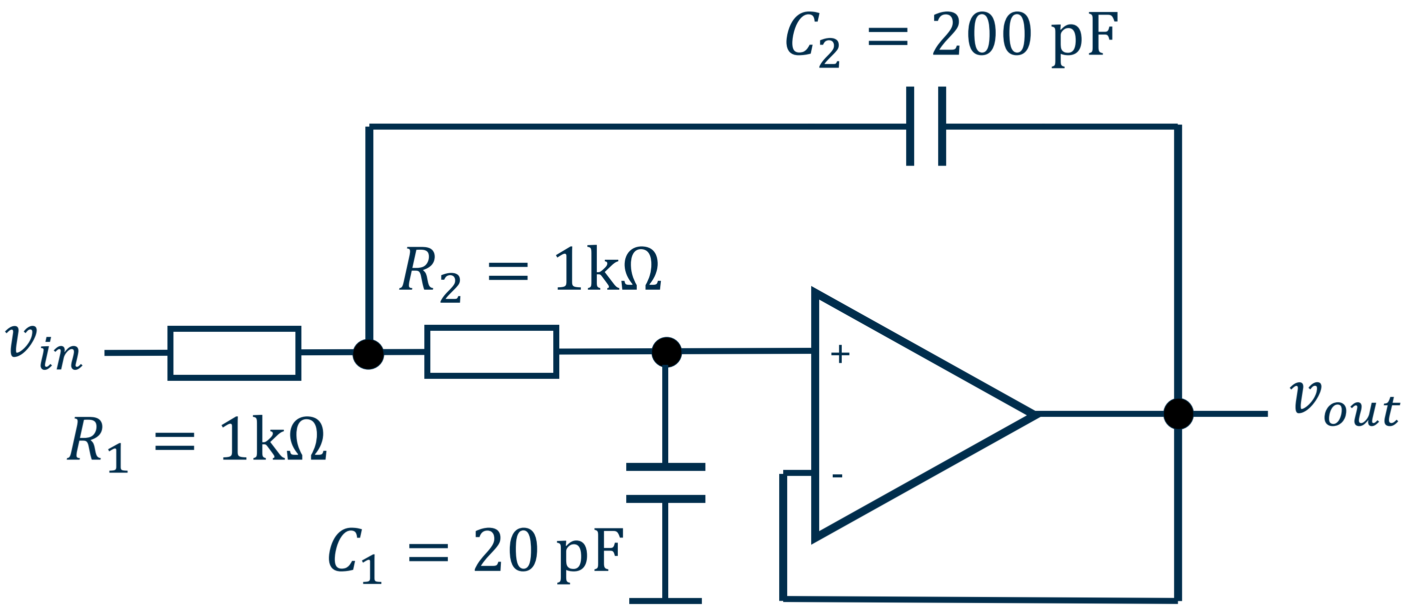
- For $\zeta = 1 \rightarrow \omega_{3dB} = 0.64\omega_0 \rightarrow f_0 = 5.05\ \mathrm{MHz}$

Sallen-Key Filter ($C_2 = 20\ \mathrm{pF}$)

- Step Response:



Sallen-Key Filter ($C_2 = 200$ pF)

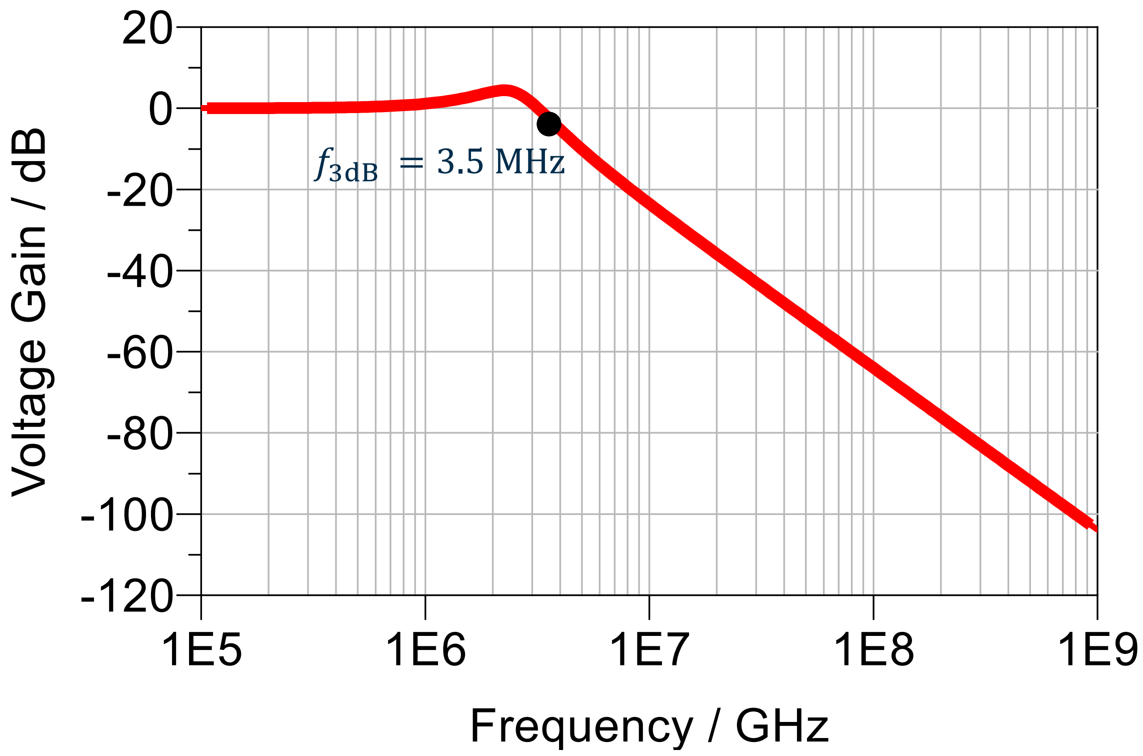


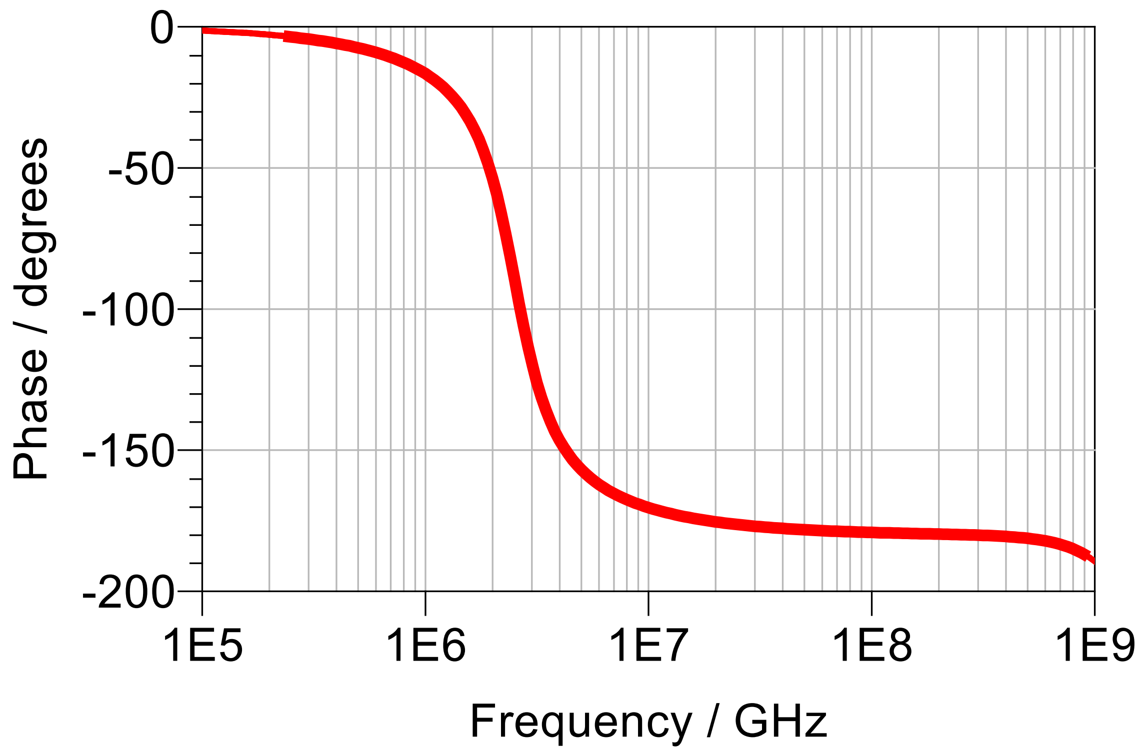
$$R_2 R_1 C_2 C_1 = 4.0 \times 10^{-15}\text{s} \rightarrow \omega_0 = 15\text{ MHz} \rightarrow = 2.5\text{ MHz}$$

$$\omega_{3dB} = 1.26\omega_0 \rightarrow f_0 = 3.15\text{ MHz}$$

$$\frac{R_1 C_1 + R_2 C_1}{R_2 R_1 C_2 C_1} \rightarrow Q = 1.5 \rightarrow \zeta = 0.33$$

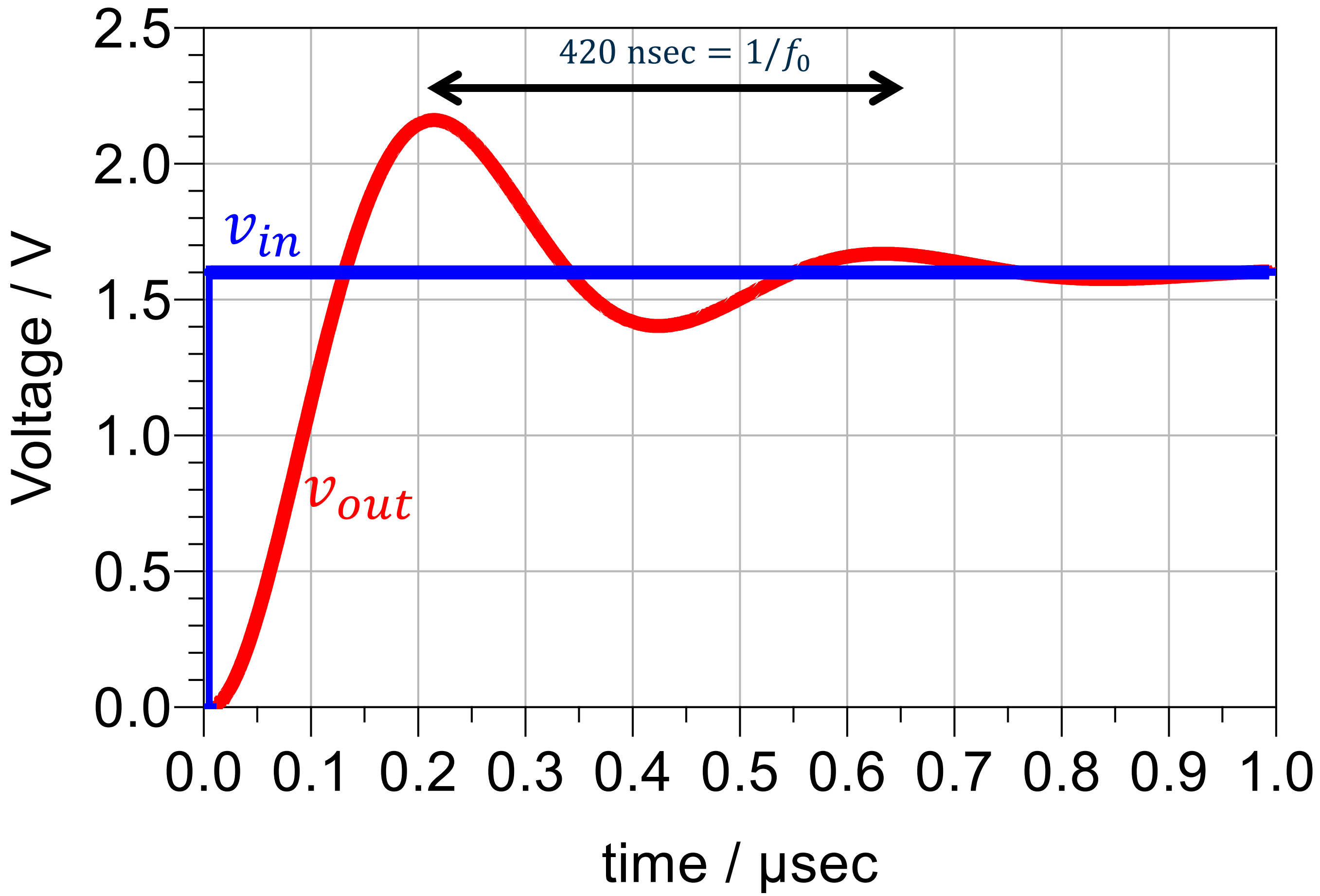
Sallen-Key Filter ($C_2 = 200\ \mathrm{pF}$)





Sallen-Key Filter $(C_2 = 200\ \mathrm{pF})$

- Step Response:



2^{nd} Order Response and Stability

- As seen, the interaction between poles (reactive components) in the circuit can lead to a second order response that can lead to an indirect relation of the 3-dB bandwidth, and ringing in the transient response
- This may sometimes be desired (e.g. Cherry-Hooper), but in other times is an undesired effect
- The complex conjugate root pair was so far assumed to have a negative real part, resulting in an exponentially decaying oscillatory behavior
- If the complex conjugate roots have positive real part, the oscillatory behavior will be exponentially growing → instability!
- Therefore, it is important to introduce a methodology that can spread the poles apart in a circuit, to avoid 2^{nd} order ringing or unstable behavior

2^{nd} Order Response and Stability

- For stability analysis, the previously introduced “Loop Gain” is the critical quantity we need to investigate

$$\frac{v_{out}}{v_{in}} = \frac{A(j\omega)}{1 + A(j\omega)F(j\omega)}$$

- For $|A(j\omega)F(j\omega)| = 1$ and $\angle A(j\omega)F(j\omega) = n180°$, the denominator of the closed loop transfer function goes to zero, resulting in infinite voltage amplification → unstable behavior (Barkhausen Criteria)
- This indicates, we can assess stability of an amplifier examining the steady-state response of the Loop Gain!

2nd Order Response and Stability

- To understand how closed loop gain is affected by feedback, we will consider a second order $A(s)$ and a constant F

$$A(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p1}}\right)}$$

- The closed loop transfer function will be:

$$H(s) = \frac{A_0 \omega_{p1} \omega_{p2}}{s^2 + \left(\omega_{p1} + \omega_{p2}\right)s + (1 + A_0 F)\omega_{p1}\omega_{p2}}$$

- with the roots

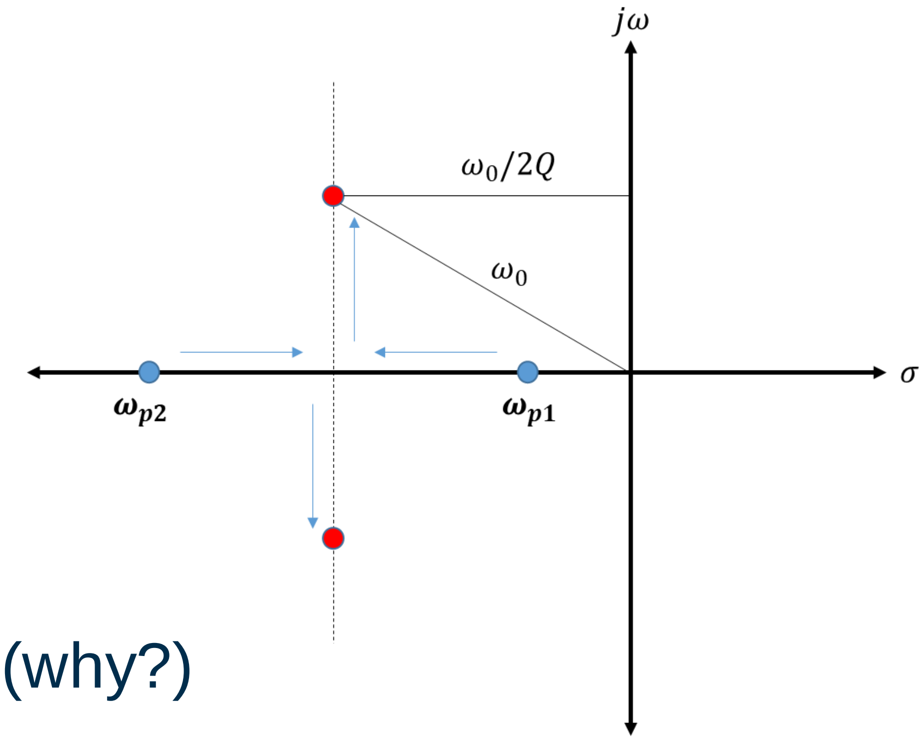
$$s_{1,2} = \frac{-(\omega_{p1} + \omega_{p2}) \mp \sqrt{\left(\omega_{p1} + \omega_{p2}\right)^2 - 4(1 + A_0 F)\omega_{p1}\omega_{p2}}}{2}$$

2^{nd} Order Response and Stability

- With no feedback ($F = 0$), the roots will be $-\omega_{p1}$ and $-\omega_{p2}$

- As, the feedback amount is increase ($F \nearrow$), the term under square-root decreases, reaching zero for critical value (having two equal real roots)

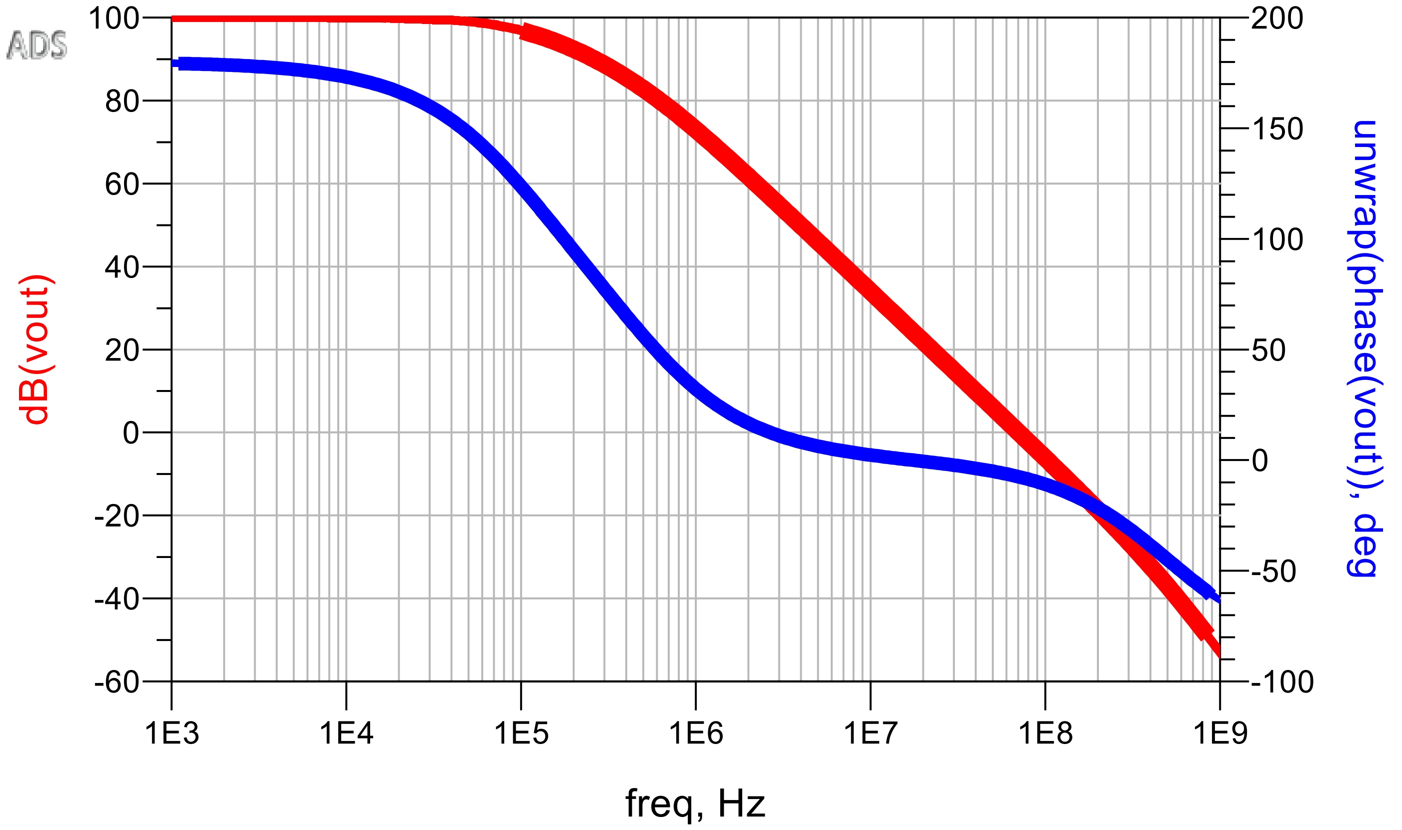
$$F_C = \frac{1}{A_0}\frac{\left(\omega_{p1} - \omega_{p2}\right)^2}{4\omega_{p1}\omega_{p2}}$$



- Becoming complex conjugate beyond this point if F is further increased
- This is the process we have observed in the Sallen-Key circuit
- A second-order amplifier cannot become unstable (why?)

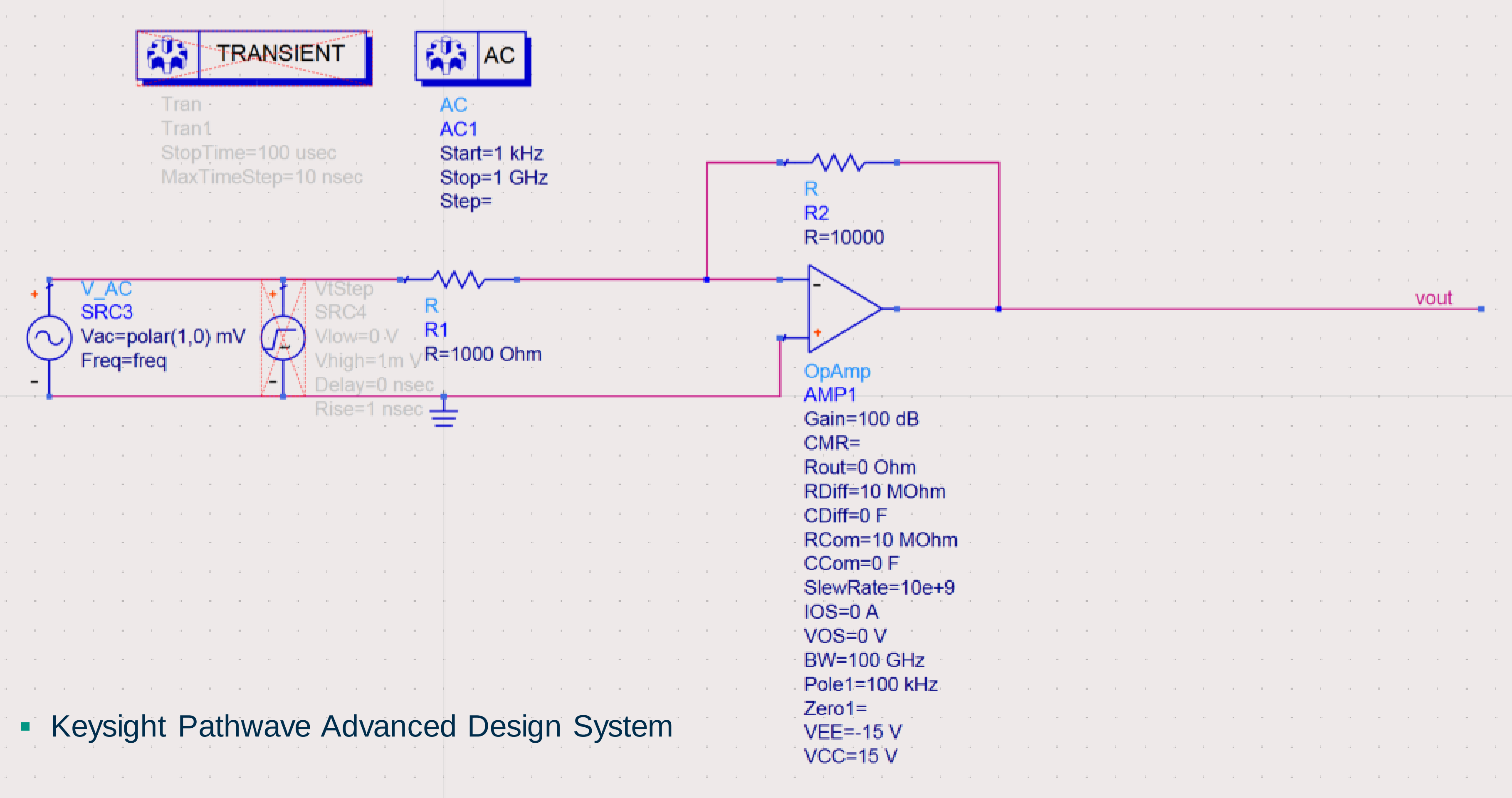
Stability and Loop Gain

- Let´s examine a third-order OPAMP with poles at 100 kHz, 500 kHz and at 500 MHz
  - open loop:



Stability and Loop Gain

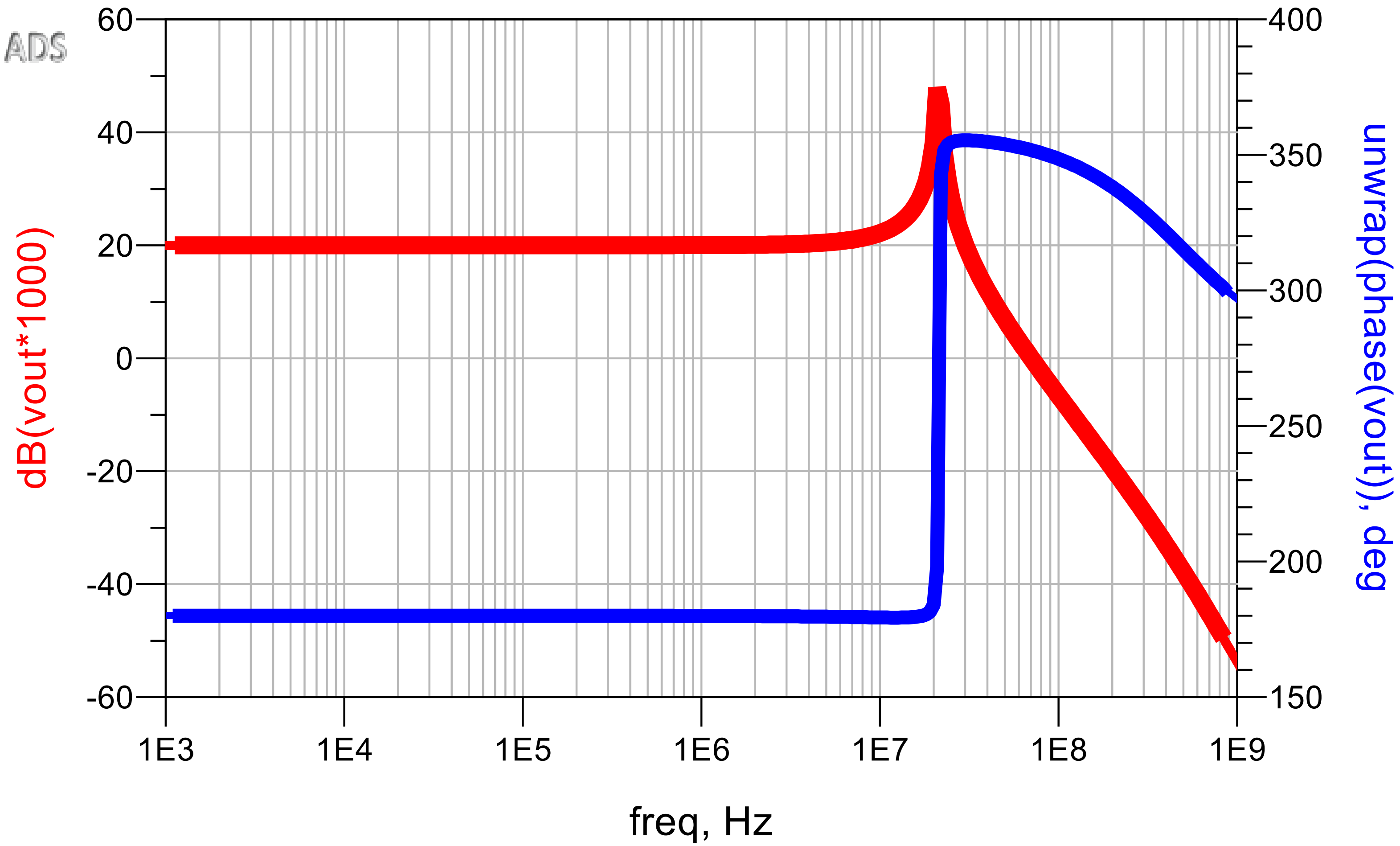
- Now, we build a feedback amplifier with a gain of 20 dB using this OPAMP



- Keysight Pathwave Advanced Design System

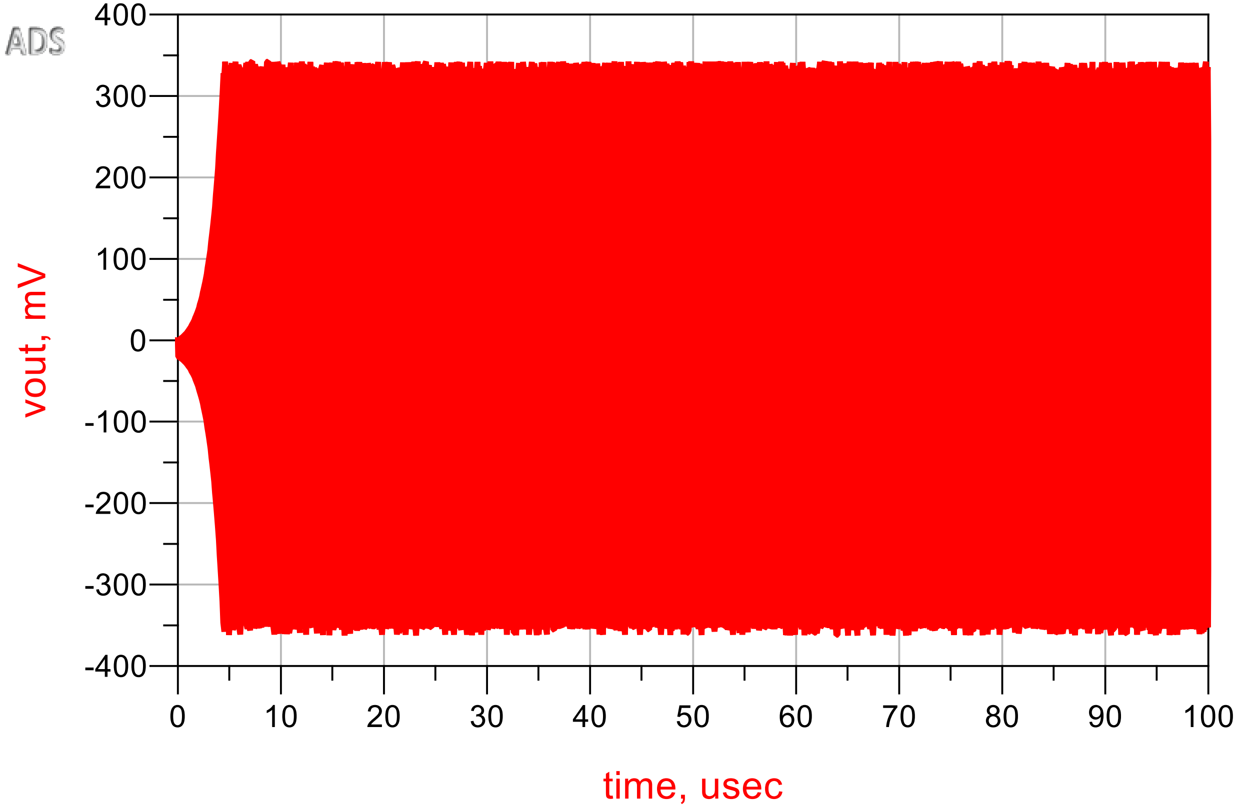
Stability and Loop Gain

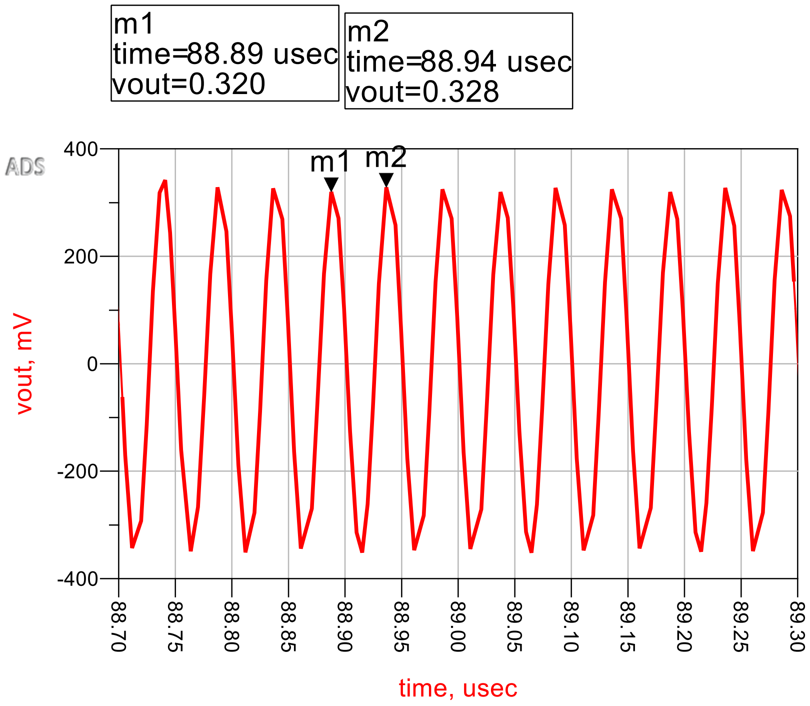
- Now, we build a feedback amplifier with a gain of 20 dB, using this OPAMP
  - closed loop:



Stability and Loop Gain

- Step Response:

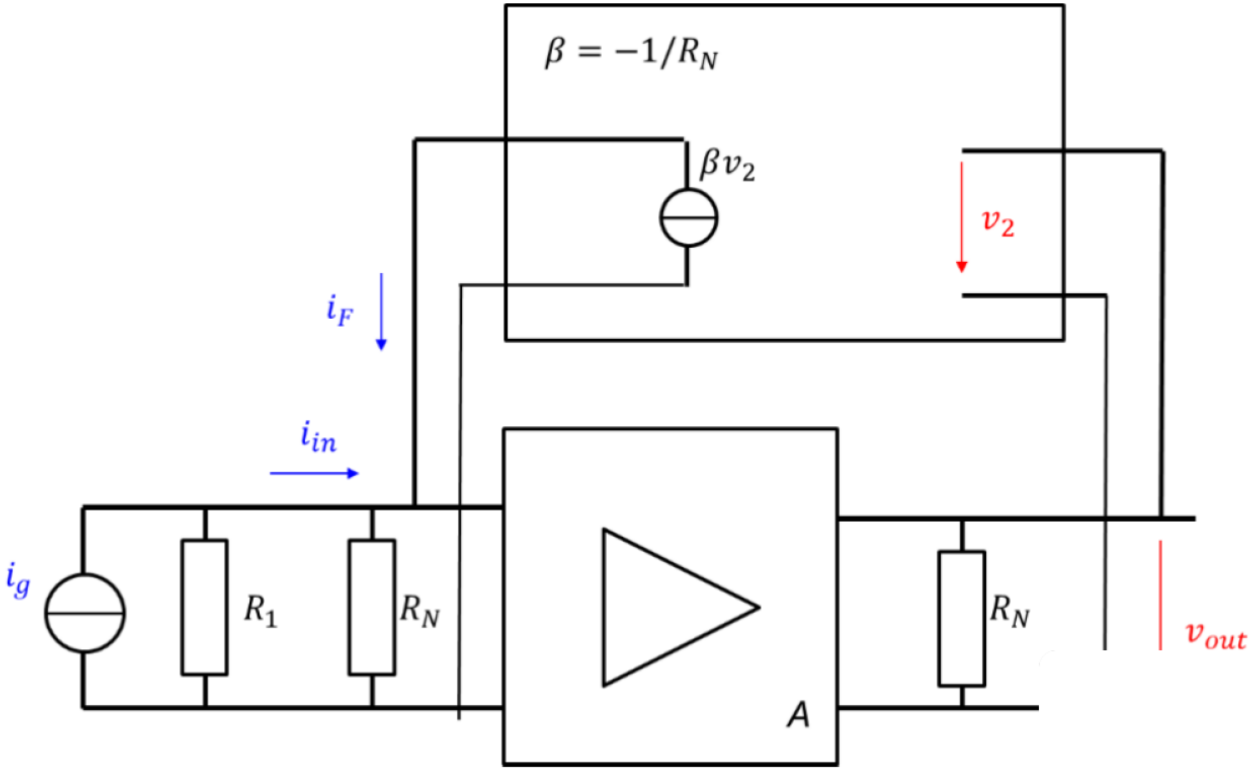




- f_0 around 20 MHz

Stability and Loop Gain

- As seen, closed loop gain does not tell us necessarilly whether the amplifier is stable or not, however if we examine the loop gain (here using classical method)



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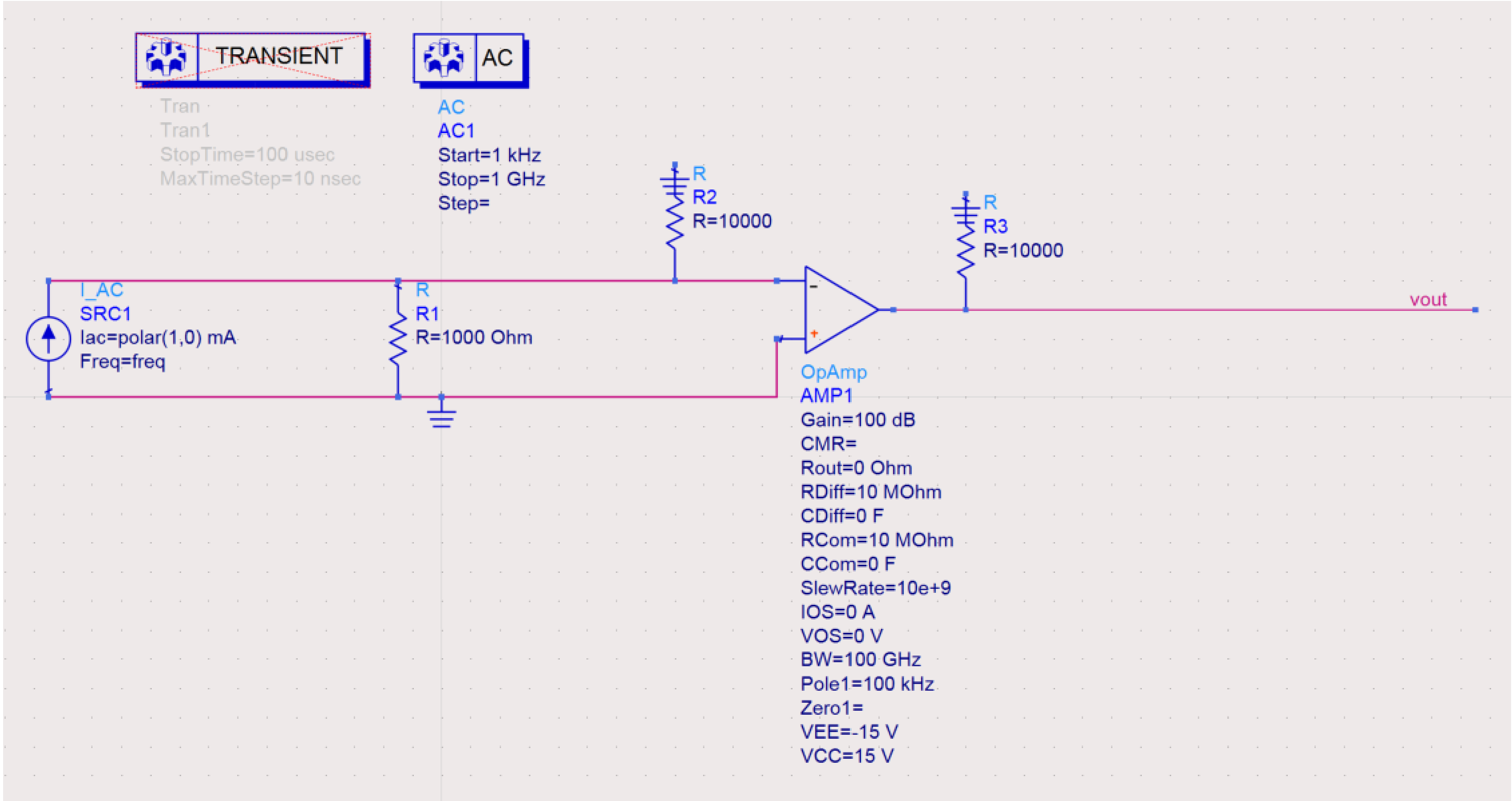
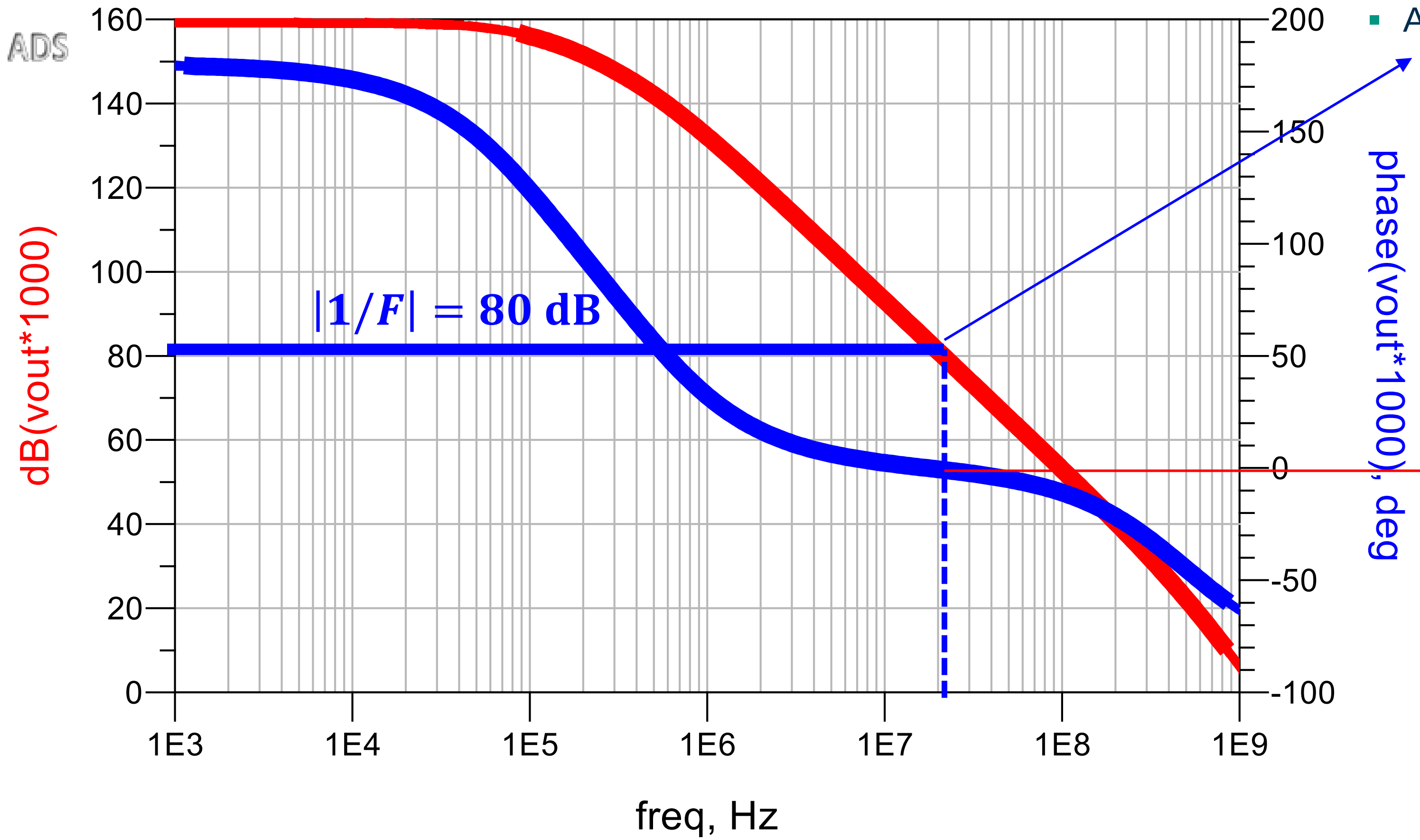


Stability and Loop Gain

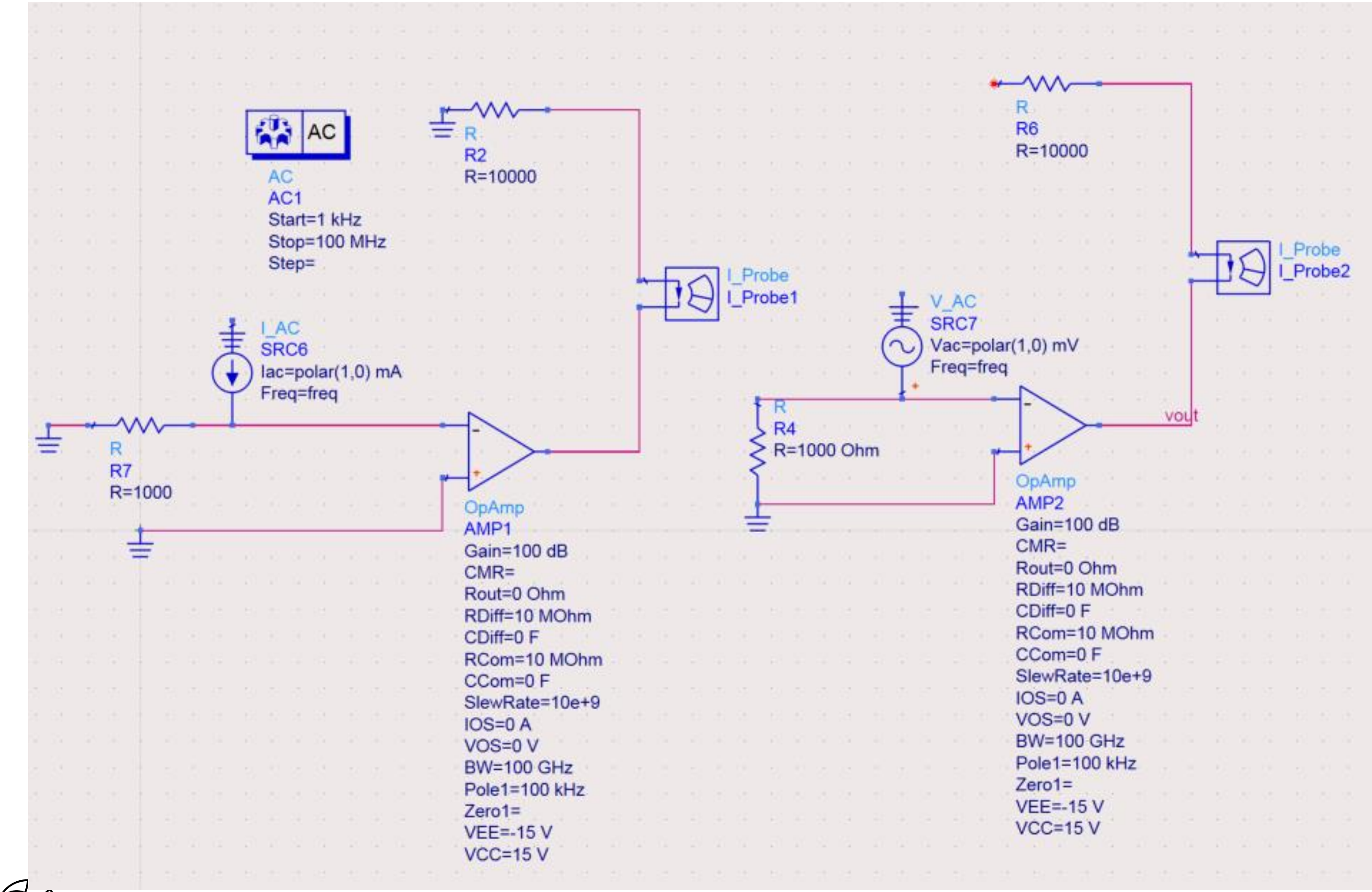
- Loop Gain (A and F seperate):



- At the intersection loop gain $|AF| = 1$
- Phase exceeds 180° by ~1°

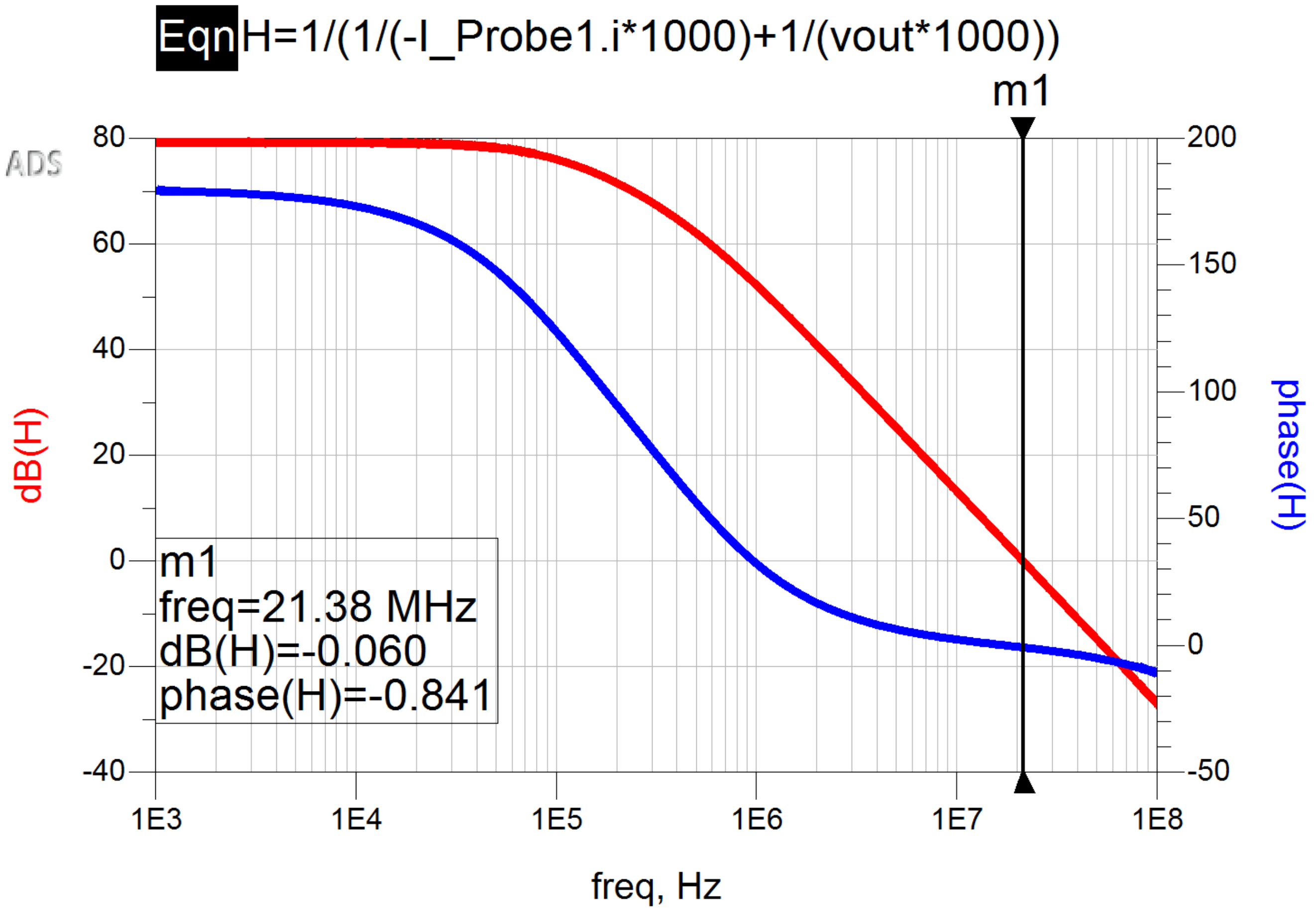
Stability and Loop Gain

- Loop gain assessed using: $\dfrac{1}{AF} = \dfrac{1}{A_{vop}} + \dfrac{1}{H_{os}}$



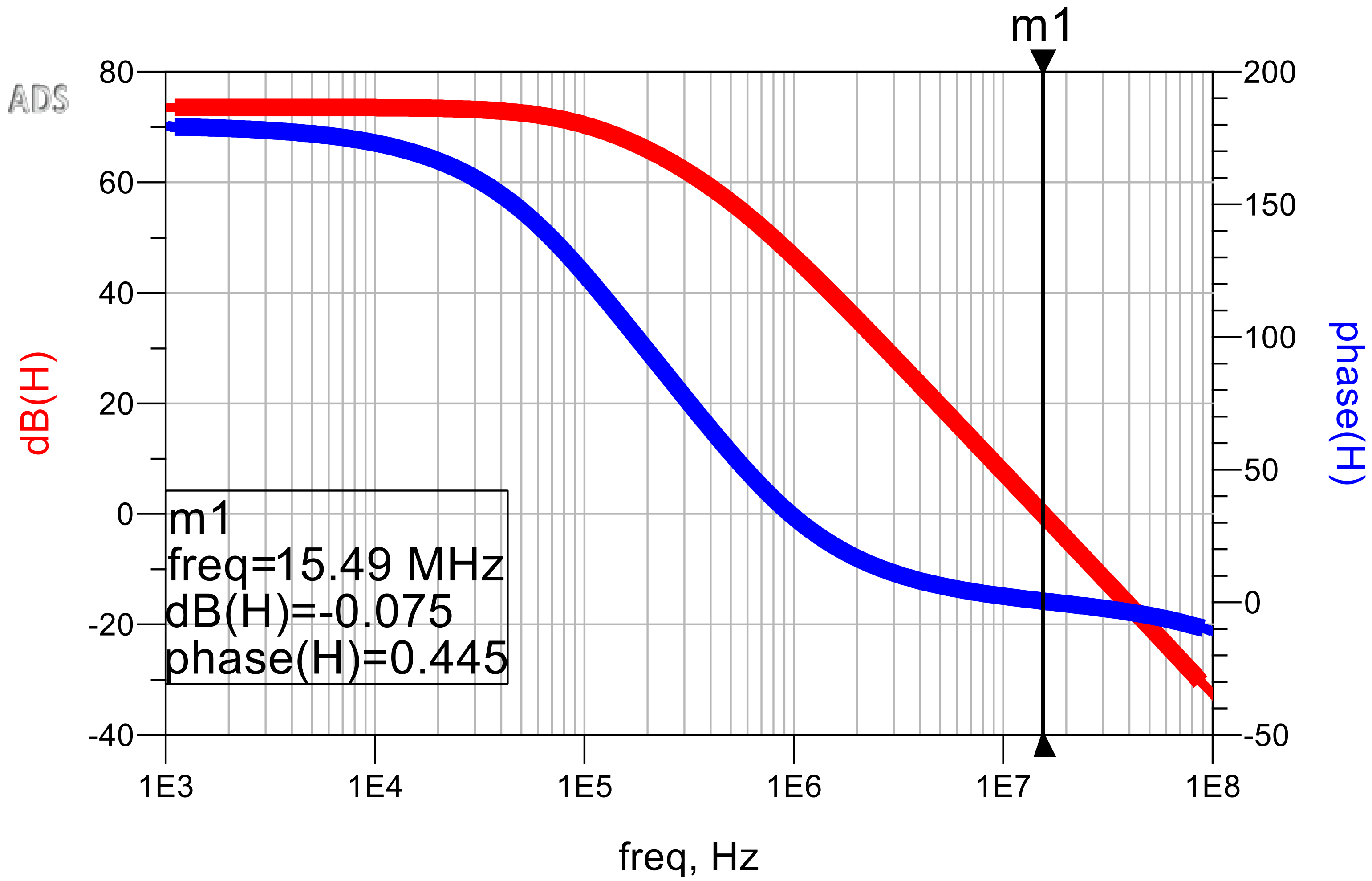
Stability and Loop Gain

- Loop gain assessed using: $\frac{1}{AF} = \frac{1}{A_{vop}} + \frac{1}{H_{os}}$



Phase Margin

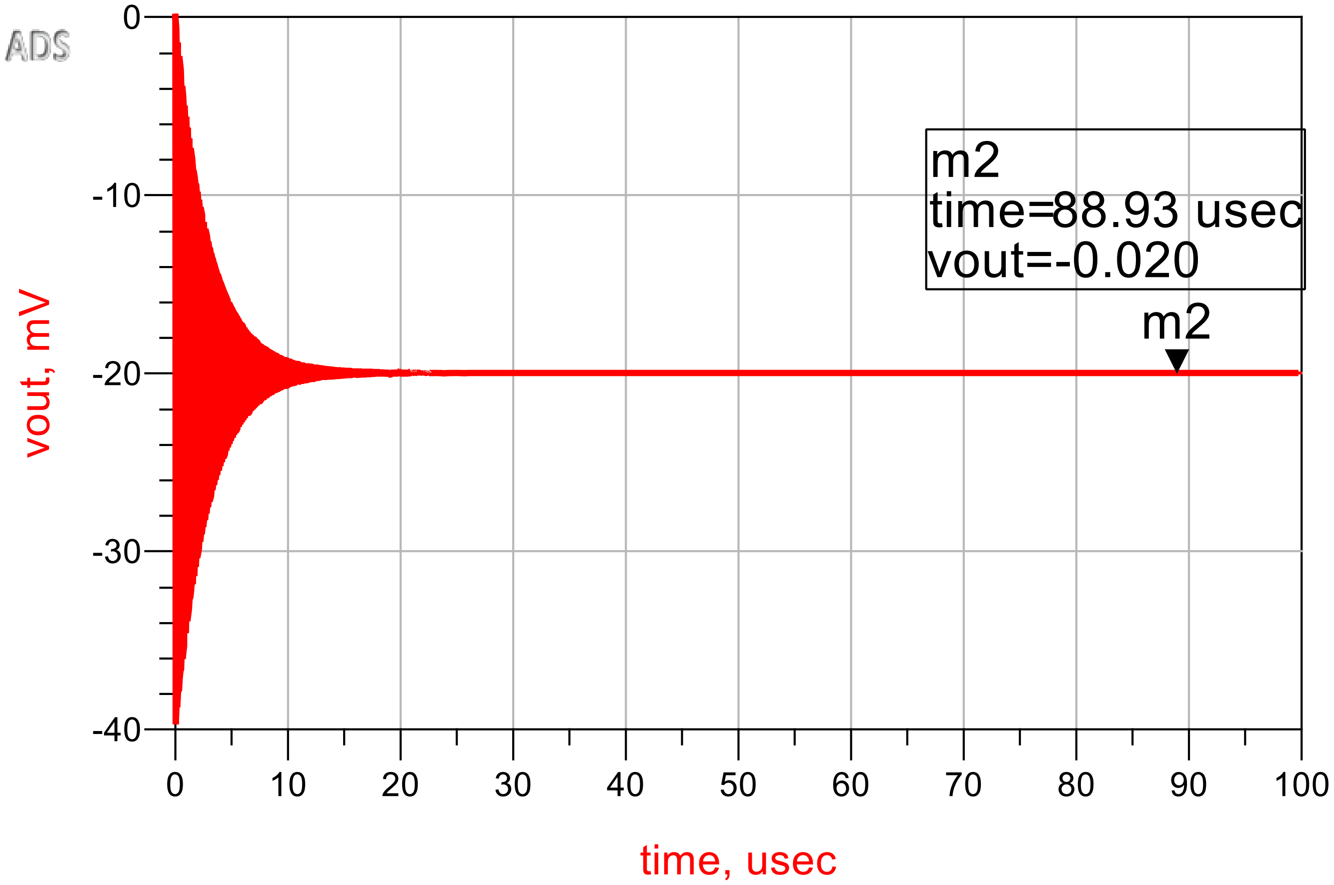
- Feedback resistance increased to $20\ \mathrm{k\Omega}$ (less feedback, closed loop gain **increased**)



- When loop gain $|AF| = 0\ \mathrm{dB}$, the phase response is 0.5° below 180° of phase shift
- This is called the **phase margin**
- The magnitude at 180° phase shift is called the **amplitude margin**
- **In this case the phase margin is only 0.5°**

Phase Margin

- Step Response (closed loop with $20\ \mathrm{k\Omega}$):



- A gain of -20 as expected ($|v_{in}| = 1\ \mathrm{mV}$)
- A significant ringing response is observed, much stronger than the actual signal amplitude
- No sustained oscillation, the oscillation decays exponentially in steady state

Phase Margin

- How much phase margin is good enough?
- We observe the closed loop transfer function in the vicinity of $|A| = 1$ at frequency ω_1

$$H(j\omega_1) = \frac{A(j\omega_1)}{1 + A(j\omega_1)F}$$

Phase Margin

- An important question is, how much phase margin is good enough?
- We observe the closed loop transfer function in the vicinity of $|A| = 1$ at frequency ω_1

$$H(j\omega_1) = \frac{A(j\omega_1)}{1 + A(j\omega_1)F} \rightarrow \frac{\frac{1}{F} e^{-j179°}}{1 + e^{-179°}}$$

- For our example with a phase margin of 1°, $A(j\omega_1)F$, will have a magnitude of 1, and a phase of -179°

Phase Margin

- An important question is, how much phase margin is good enough?
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$$H(j\omega_1) = \frac{A(j\omega_1)}{1 + A(j\omega_1)F} \rightarrow \frac{\frac{1}{F}e^{-j179°}}{1 + e^{-179°}} \rightarrow \frac{1}{F}\left(\frac{-0.997 - j0.0707}{0.003 - j0.0707}\right) \rightarrow |H| = \frac{14.1}{F}$$

- For our example with a phase margin of 1°, $A(j\omega_1)F$, will have a magnitude of 1, and a phase of -179°
- Compared to asymptotic response of $\frac{1}{F}$, we will observe a peaking of 24 dB at ω_1

Phase Margin

- An important question is, how much phase margin is good enough?
- We observe the closed loop transfer function in the vicinity of $|A| = 1$ at frequency ω_1

$$H(j\omega_1) = \frac{A(j\omega_1)}{1 + A(j\omega_1)F} \rightarrow \frac{\frac{1}{F} e^{-j179°}}{1 + e^{-179°}} \rightarrow \frac{1}{F}\left(\frac{-0.997 - j0.0707}{0.003 - j0.0707}\right) \rightarrow |H| = \frac{14.1}{F}$$

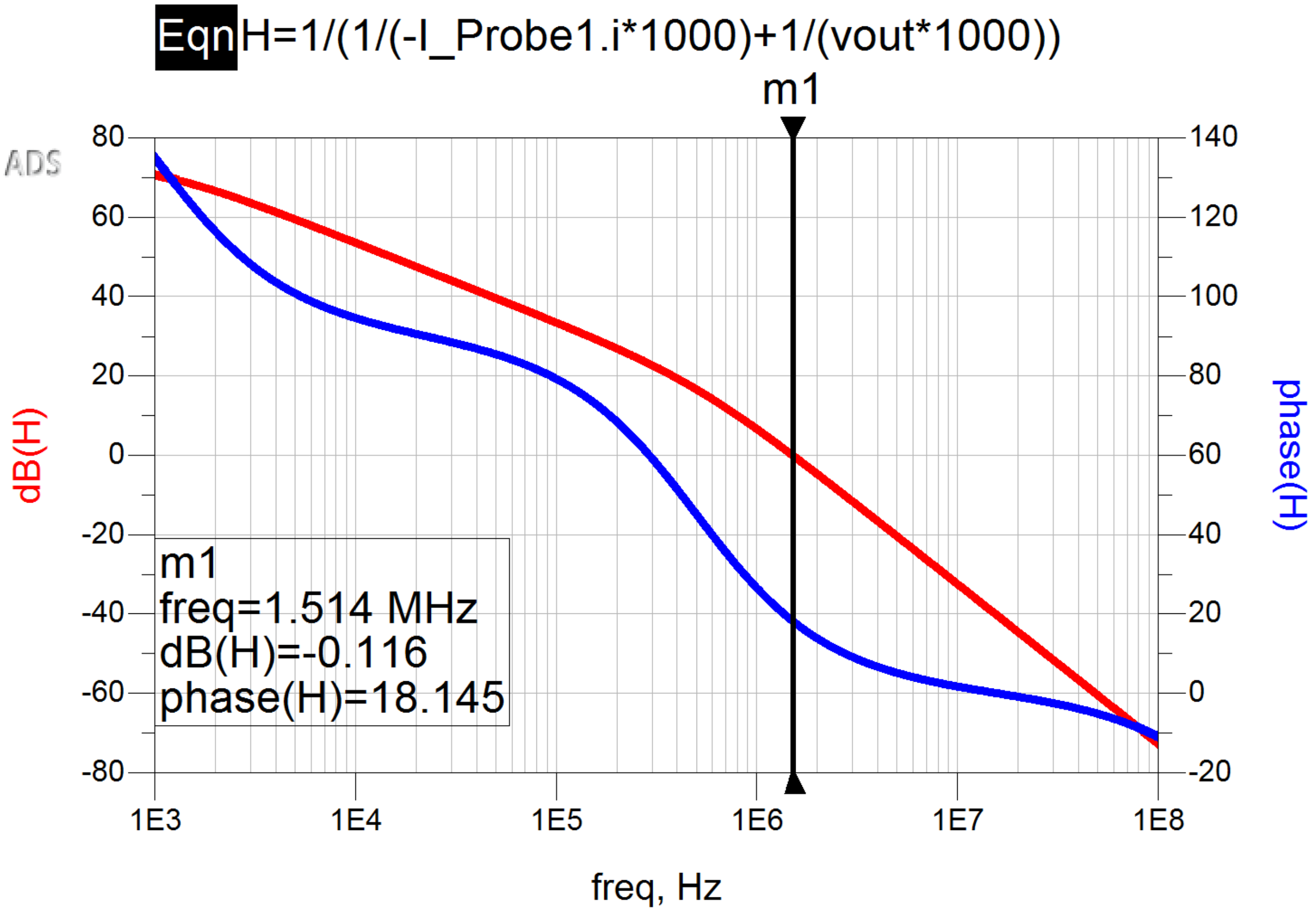
- For our example with a phase margin of 1°, $A(j\omega_1)F$, will have a magnitude of 1, and a phase of -179°
- Compared to asymptotic response of $\frac{1}{F}$, we will observe a peaking of 24 dB at ω_1
- At a **phase margin of 60°,** ringing behavior can be considered negligible
- At a phase margin of 45°, the peaking is around 2.2 dB, with still noticeable ringing

Frequency Compensation

- In order to increase phase margin, ideally we would like to build a circuit with as low number of poles as possible → reduce number of stages & reactive components
- If this is not possible, a typical approach would be to reduce the lowest pole frequency (making it dominant) so that the magnitude decays faster before reaching a phase shift of 180° → this means gain-bandwidth product will be lower
- In OPAMP designs this is a standard method, as typically multiple stages are required to reach sufficient gain, and frequency compensation is applied to create a dominant pole frequency

Frequency Compensation

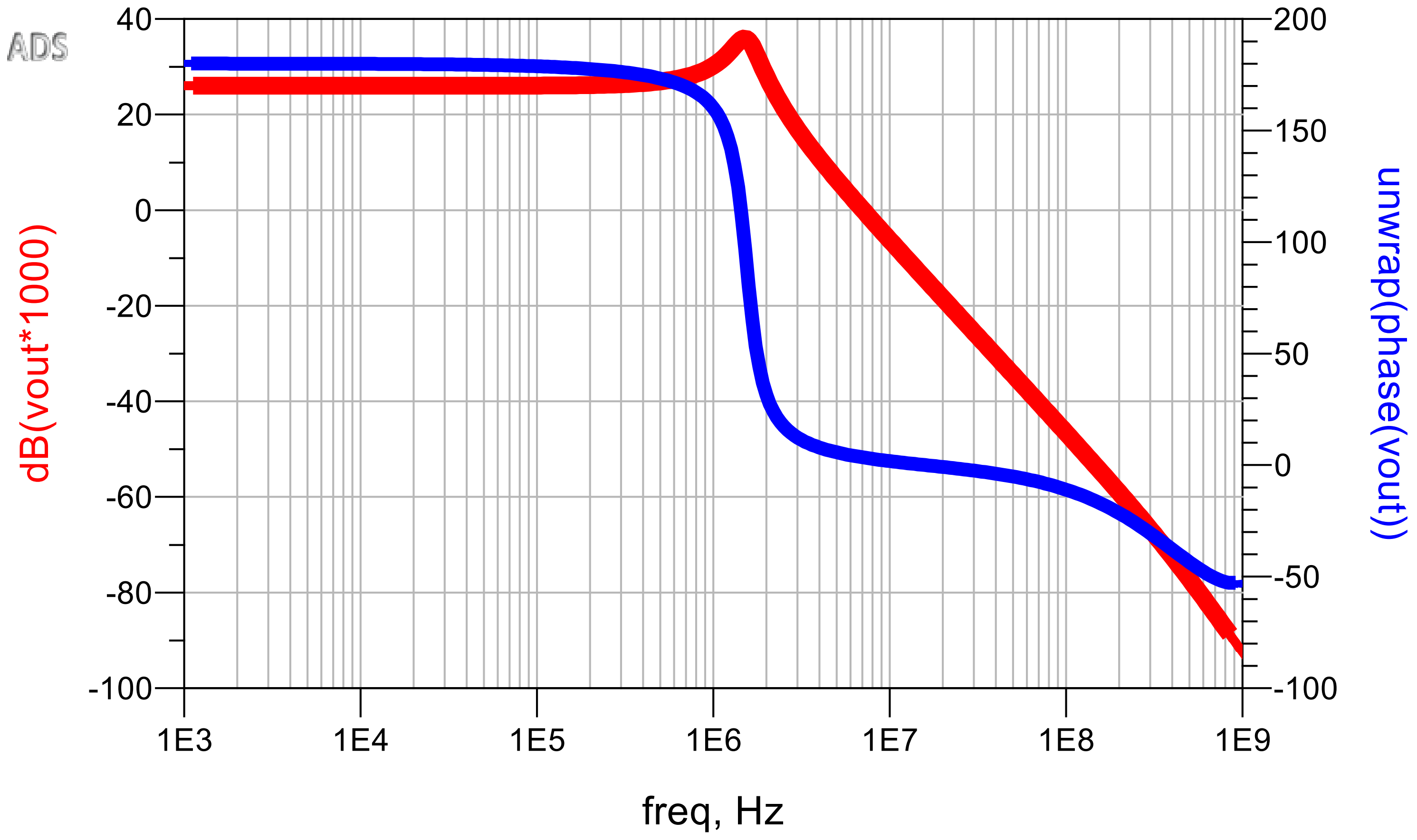
- In our previous example, first pole frequency is reduced to 1 kHz (from 100 kHz)



- **In this case the phase margin is almost 20°**
- **How can a pole frequency be reduced?**

Frequency Compensation

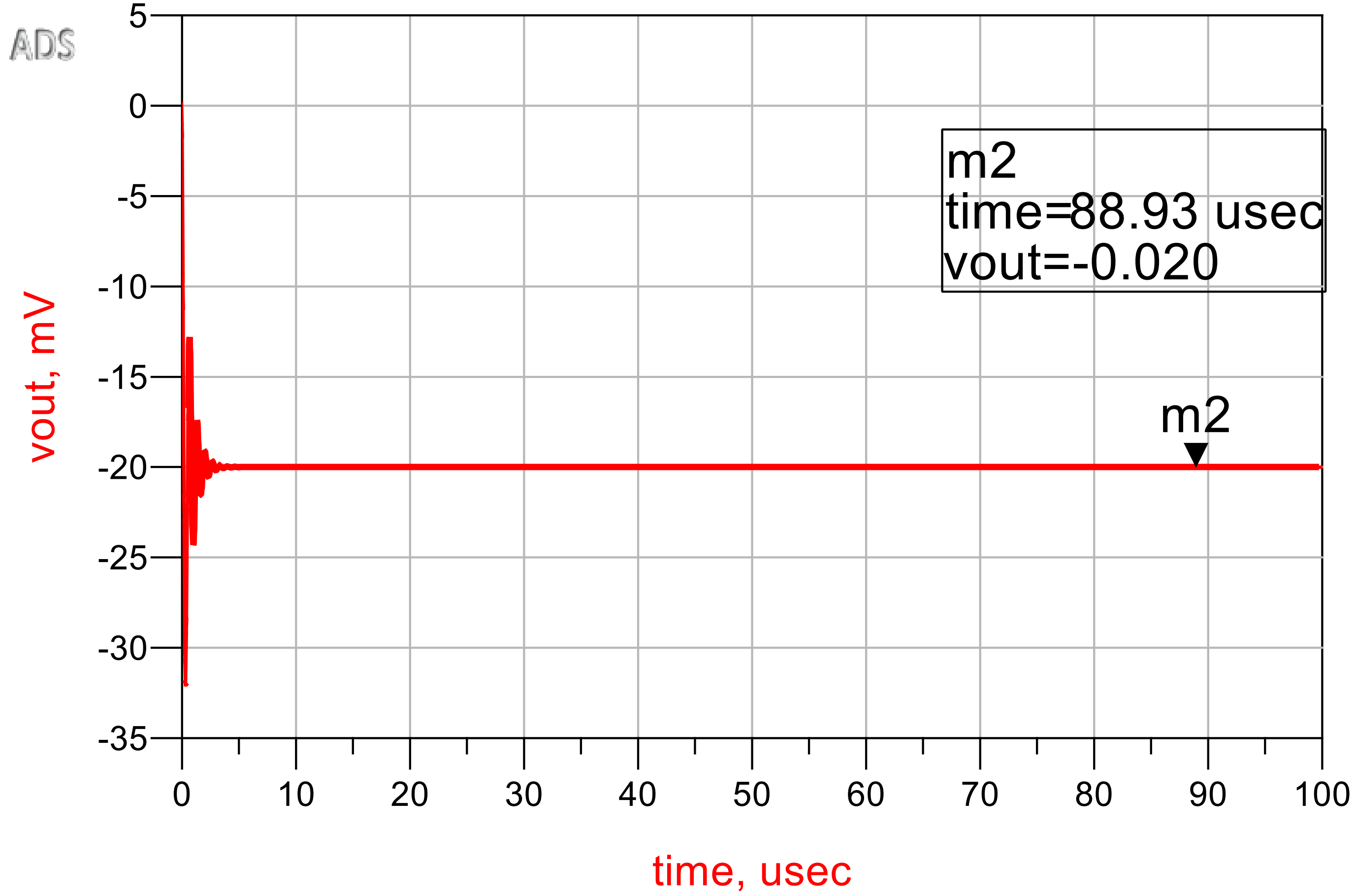
- In closed loop:



- **Peaking is reduced to 9 dB**

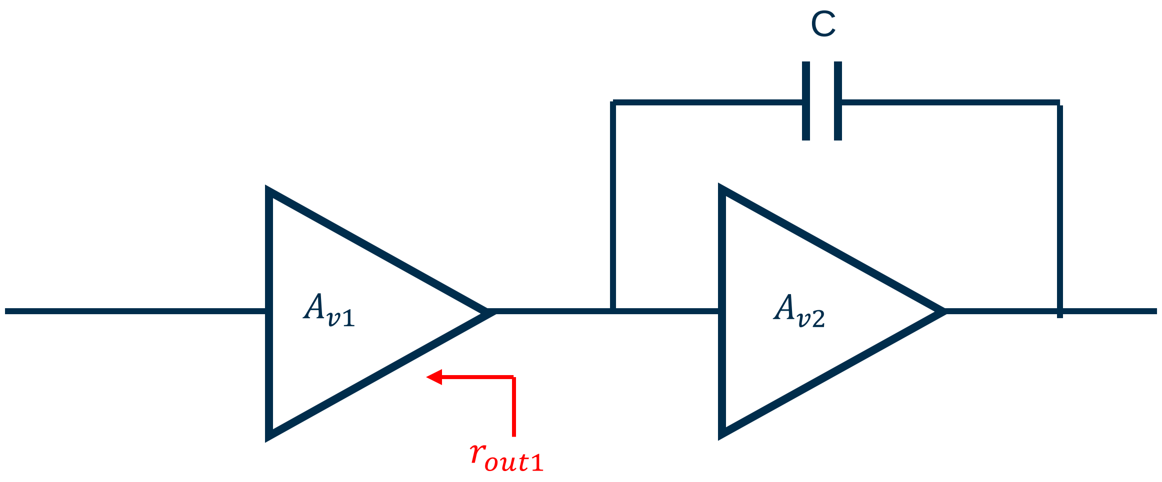
Frequency Compensation

- Ringing is reduced:



Miller Compensation

- In multi-stage designs, we typically see this kind of configuration:



- The capacitance in Feedback will be enhanced my Miller effect:
- $(C \times (1 + A_{v2}))$, forming a time constant with r_{out1} (which is high), and lower the pole frequency at the input of the second stage significantly, which is typically the lowest pole frequency in the chain